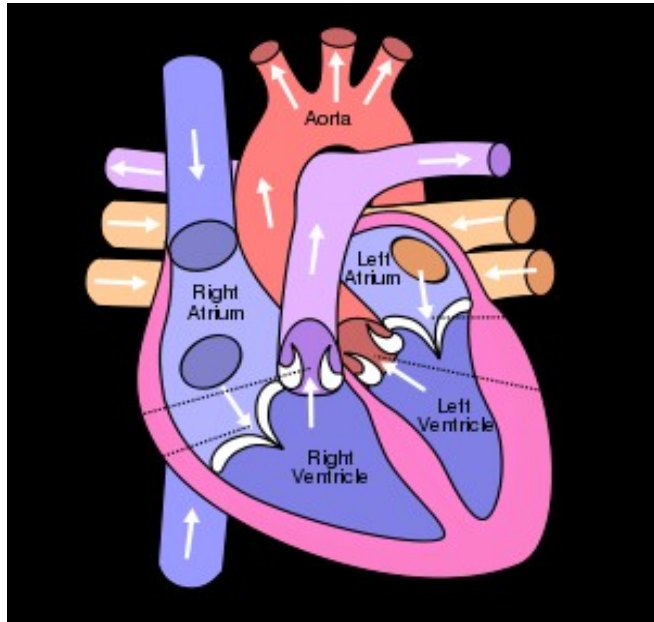




CARDIOVASCULAR SYSTEM

Objectives

- At the end of this lesson the students are expected to:
 1. List functions of cardiovascular system.
 2. Explain the components of CVS.
 3. Discuss blood flow through the heart.
 4. Enumerate the functions of the lymphatic system.
 5. List the CVS abnormalities.

Outlines

- Introduction to CVS.
- Components of CVS.
- Anatomy - Physiology of CVS.
- Blood flow through the heart & the body.
- Blood vessels types.
- Abnormalities of CVS.
- Lymphatic system.

Introduction

- **Heart** = **life-giving, ever-beating muscle** in your chest. From inside the **womb** until **death**, the thump goes on.
- The heart for the average human will **contract** about **3 billion times; never resting, never stopping** to take a break except for a fraction of a second between beats.
- At **80 years of age**, a person's heart will continue to beat an average of **100,000 times a day**.

Introduction...

- Many believe that the heart is **the first organ** to become functional.
- Within weeks of conception the **heart starts its mission** of supplying the body with nutrients even though the **embryo is no bigger than a capital letter on this page**.
- The primary function of the **heart** is to pump blood through the:
 - arteries, capillaries, & veins.

Introduction...

- There are an estimated **96,000km** of **blood vessels** throughout an adult body.
 - ☞ The total length of **capillaries** in an average adult human is approximately **42,000 km**.
- **Blood** transports:
 - Oxygen & nutrients.
 - Disease causing viruses & bacteria.
 - Hormones.
- **Heart** = the **pump** that keeps blood circulating properly.

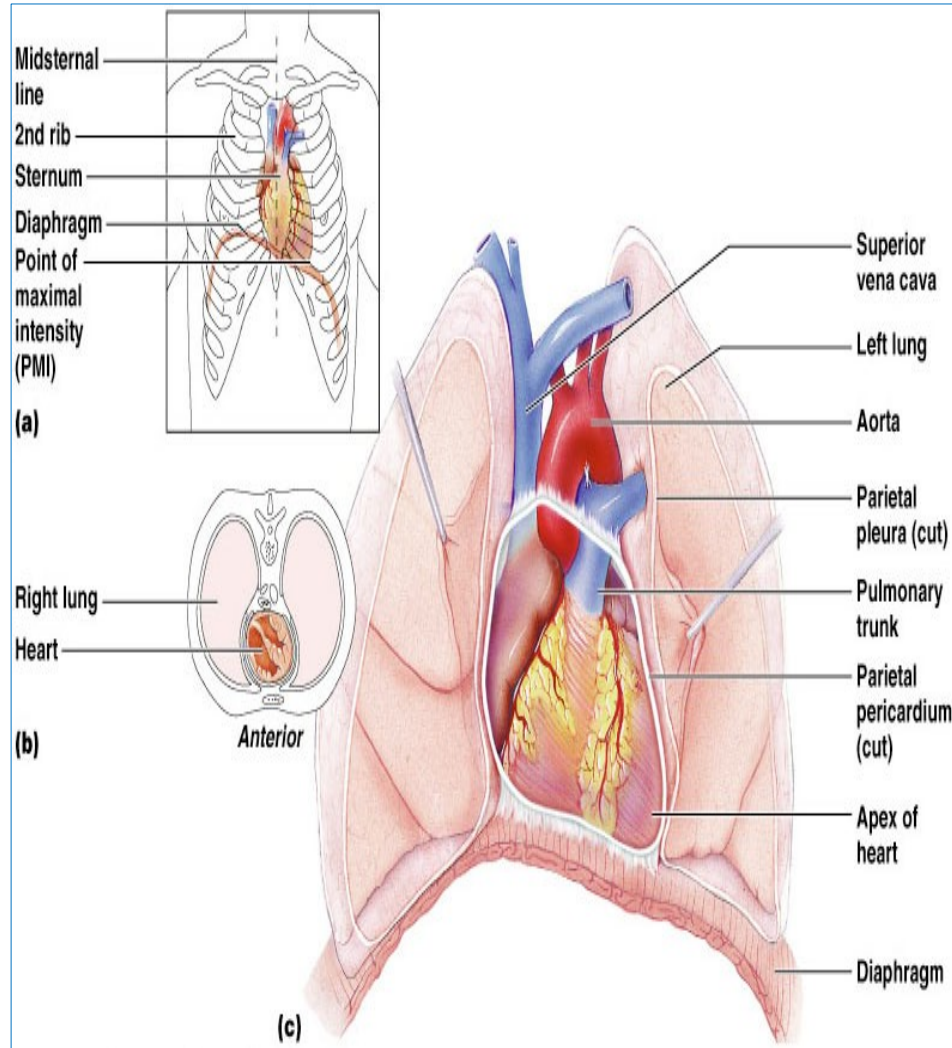
Basic components of the CVS

- CVS (cardiovascular system) consists of 3 basic components:
 1. **Heart** serve as **pump** that impart pressure on blood to flow in the tissue.
 2. **Blood vessels** serve as the **passage** through which blood is distributed to the body & subsequently returned to the heart.
 3. **Blood** serves as the **transport medium** in transported materials dissolved or suspended.

The Heart

- Hollow, muscular organ about the size of a fist.
- It is responsible for pumping blood through the blood vessels by repeated, rhythmic contractions.
- Is composed of **cardiac muscle**, an **involuntary** muscle tissue that is found only within this organ.
- The term "**cardiac**" (as in cardiology) means "**related to the heart**" & comes from the Greek word kardia, for "**heart**."

The Heart...



- 4-chambered, double pump.
 - Located in the thoracic cavity between the lungs.
1. Pulmonary (lungs) circulation.
 2. Systemic (the body) circulation.

The Heart...

- The cardiac muscle is **self-exciting**.
 - meaning **it has its own conduction system**.
- This is in contrast with **skeletal muscle**, which requires either conscious or reflex nervous stimuli.
- The heart's rhythmic contractions occur spontaneously.
- Although the **frequency** or **heart rate** can be changed by **nervous** or **hormonal** influence such as **exercise** or the **perception of danger**.

Pericardium

- The heart is **covered** by the membrane **pericardium** which is subdivided into a visceral layer & a parietal layer.
 - The **visceral pericardium** is the **outer layer of the heart** itself and is also known as the **epicardium**.
- The space between the 2 layers of the pericardium is the **pericardial cavity**. It normally contains 10-20mL of pericardial **fluid** secreted by the membranes.
 - The presence of this fluid **reduces the friction** between the beating heart and the surrounding tissues.
- A variety of pathogens may infect the pericardium, causing **pericarditis**.

Basic Heart Anatomy

- Basically, the heart is an organ with **4** hollow **chambers** separated by wall of tissue having **3** distinct **layers**.

1. Endocardium

- a unique type of thin epithelial tissue that forms **innermost layer** of the heart.

2. Myocardium

- **thickest layer** (the bulk of the heart wall), consists of cardiac muscle responsible for the heart to **contraction**.

3. Epicardium

- **thin, external membrane** around the heart.

Basic Heart...

- The basic job of the heart is **pumping blood** through **2 separate circuits**.

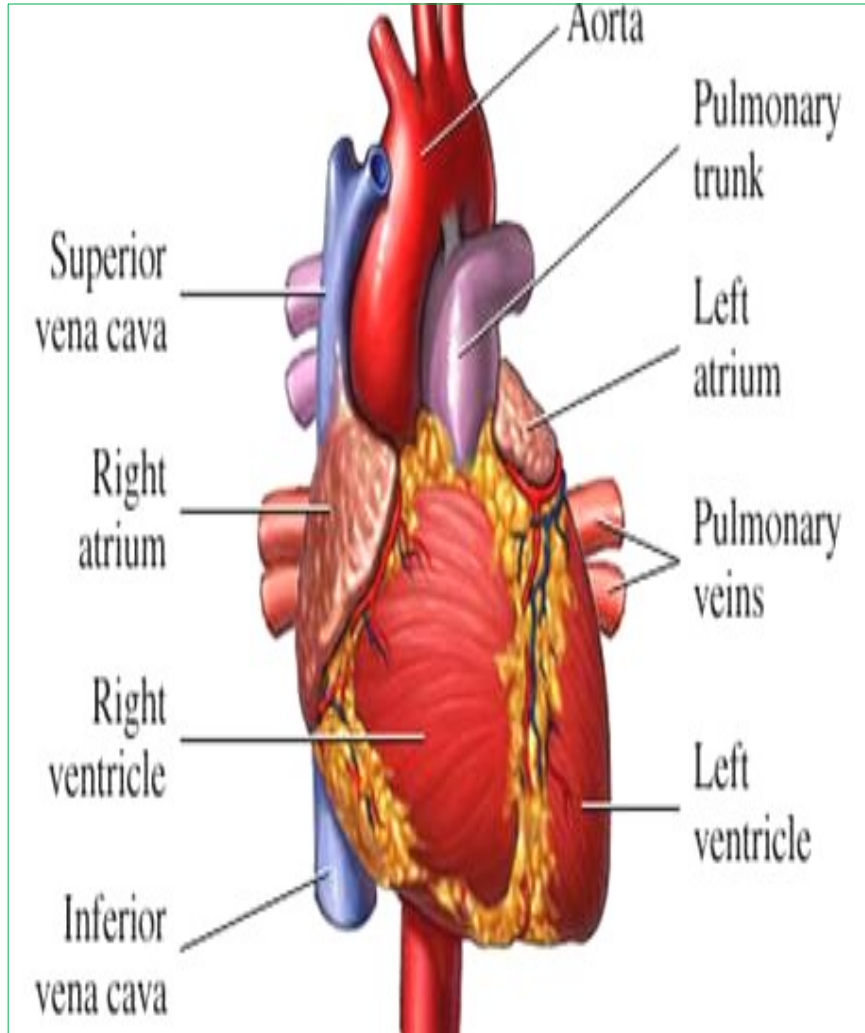
1. Pulmonary circulation

- from **right ventricle** to the gas exchange surfaces of the **lungs** & return to the **left atrium**.

2. Systemic Circulation

- from the **left ventricle** to the **tissues** in the rest of the body and return to the **right atrium**.

Heart Chambers



□ The heart contains 4 muscular chambers.

1. Right atrium:

- receives **deoxygenated blood** from the systemic circuit & passes it into RV.

2. Right ventricle:

- **discharges** deoxygenated blood into the **pulmonary circuit**.

Heart Chambers

3. Left atrium:

- receives **oxygenated blood** from the pulmonary circuit and passes it into LV.

4. Left ventricle:

- **discharges** oxygenated blood into the **systemic circuit**.
- The 2 atria are superior to the 2 ventricles.
- **Atria** have **thin, flaccid walls** corresponding to their **light workload**.

Heart Chambers...

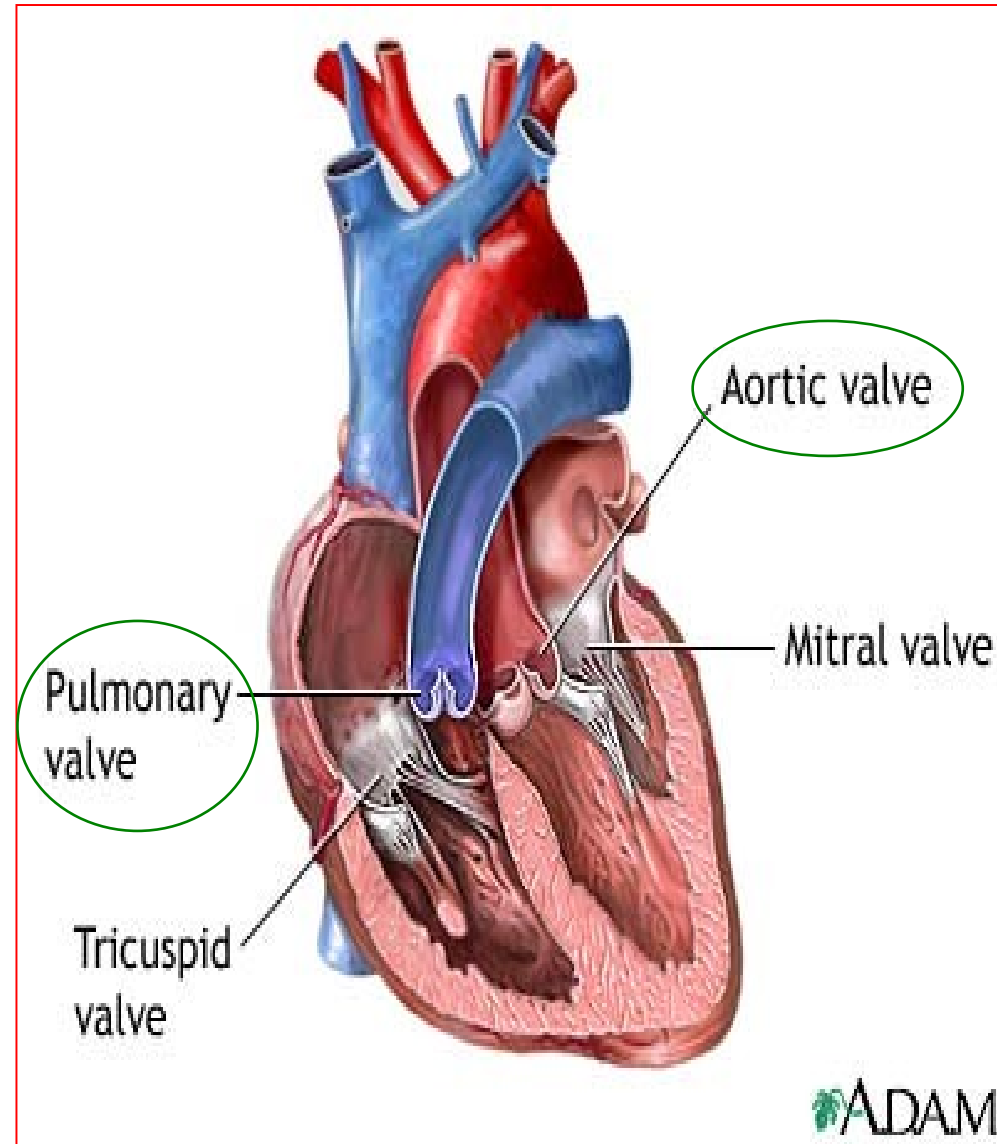
- **Atria** do not need much muscle b/c they primarily serve as **recipients** for the blood returning from the systemic & pulmonary circulation.
- Most of this blood flows into the ventricles due to **pressure difference** during ventricular relaxation.
 - They need only to contract to send a small amount of blood to the nearby ventricles.
- The right and left atrium are separated by the **interatrial septum**.

Heart Chambers...

- **Ventricles** have **thick walls**.
 - The **left ventricle** is 2-4x as thick as the right ventricle because of its **greater workload**.
- The LV and RV pump **EQUAL** amounts of blood, but the **LV** has to pump blood to the whole body & this necessitates more muscle.
- The thick **interventricular septum** separates the left and right ventricles

Valves

- Consist of 2-3 flaps of connective tissue covered by endothelium.
- The heart contains two pairs (4) valves necessary to ensure **1-way flow**.



Valves...

1. The 2 **atrioventricular valves** allow blood to flow from the **atria** to the **ventricles** and prevent the reverse direction.
 - Tricuspid & Bicuspid/mitral valve.
2. The 2 **semilunar valves** allow blood to leave the **ventricles** & enter the **pulmonary artery** & **aorta**, but these valves prevent blood return from the arteries to the ventricles.
 - Aortic & Pulmonary valve.

Atrioventricular valves

- They are found between the **atria** and **ventricles**.
- The **tricuspid valve** is found between the right atrium and the right ventricle.
 - It's given its name because it contains 3 valve flaps/cusps.
- The valve between the left atrium and the left ventricle has 2 flaps and is thus known as the **bicuspid valve**.
- The bicuspid valve is also frequently referred to as the **mitral valve**.

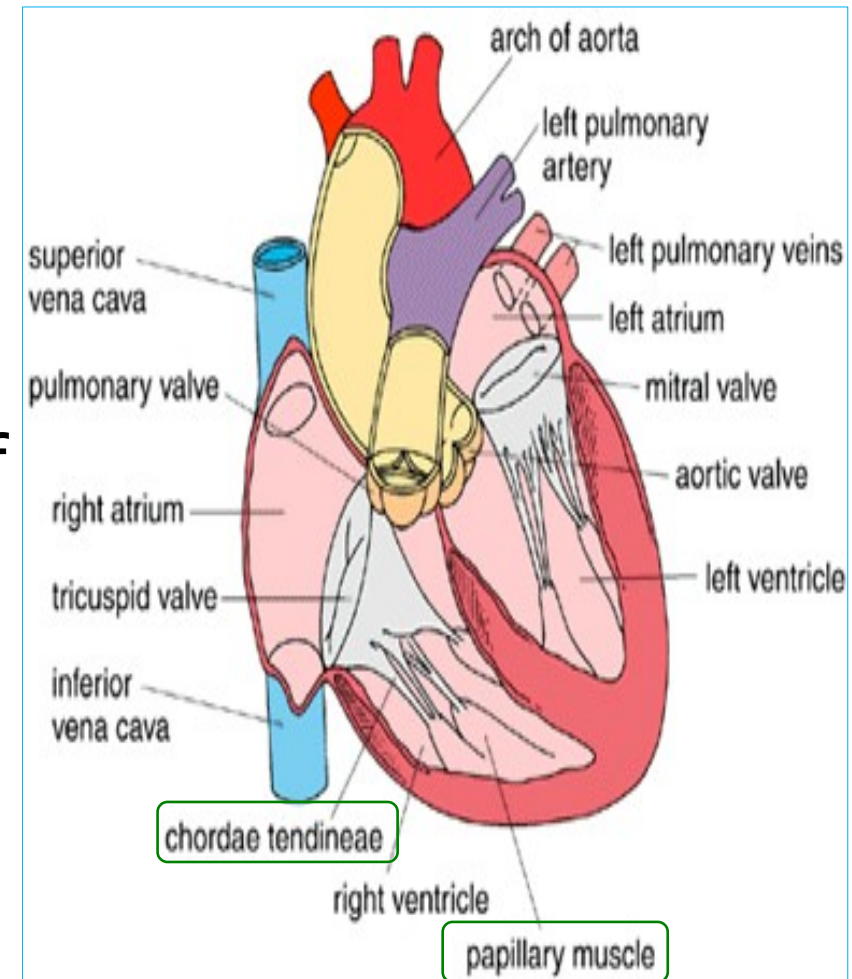
AV valves...

- **Chordae tendineae**

- Flaps of AV-valves attached to the strings of collagen.

- **Papillary muscles**

- Conical muscles that bulge from the ventricle floor.

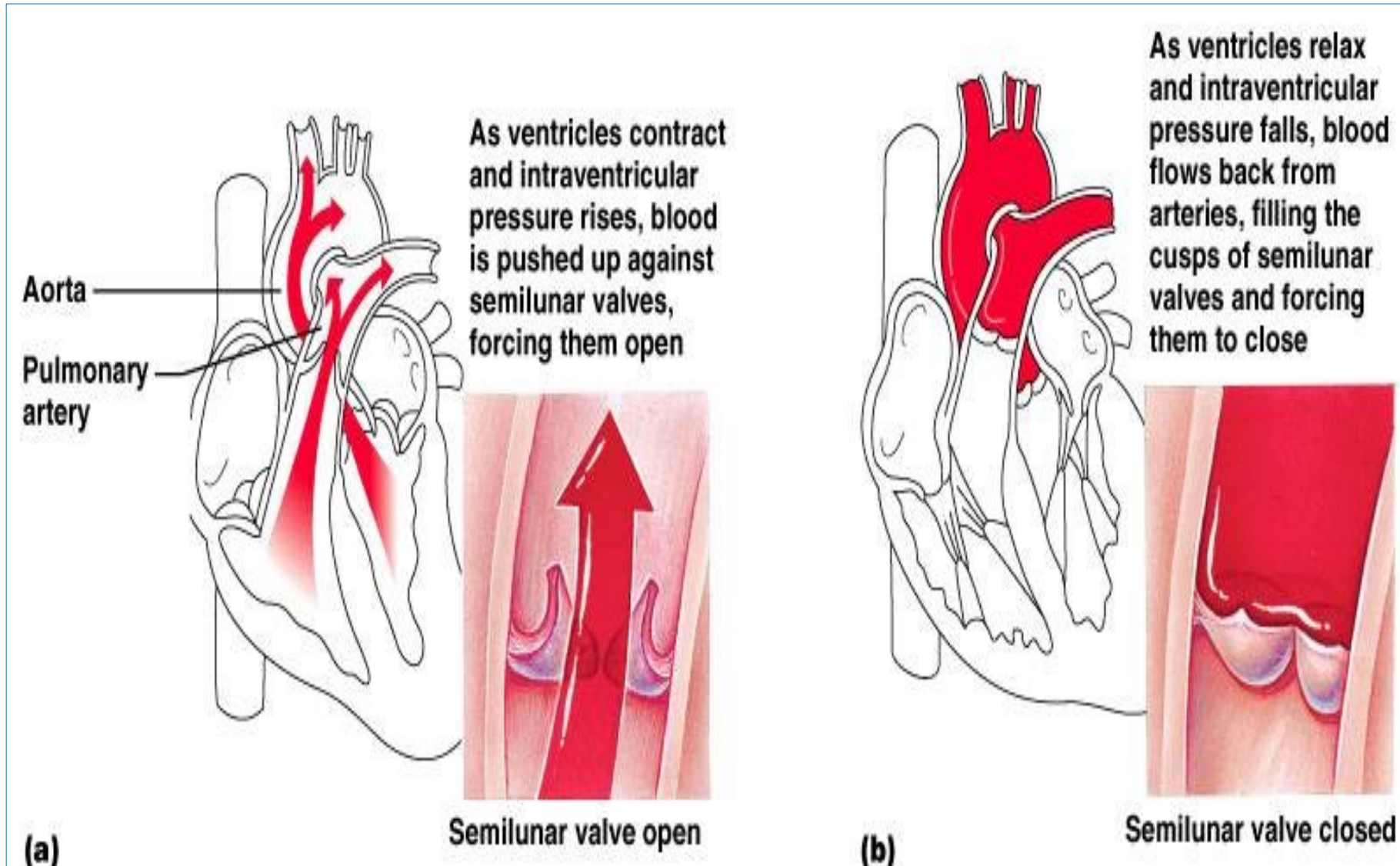


- The papillary muscles & chordae tendineae **prevent** the valve flaps from flipping upward into the atria (**prolapse**) during **high pressure ventricular contraction**.

Semilunar valves

- Located at junction of **ventricles** with **aorta** & **pulmonary trunk**.
- **Prevent backflow of blood** from aorta & pulmonary trunk into the left & right ventricles respectively.
 - **Pulmonary semilunar valve**
 - prevents backflow from the pulmonary trunk into the right ventricle.
 - **Aortic semilunar valve**
 - prevents backflow from the aorta into the left ventricle.

Semilunar valves...

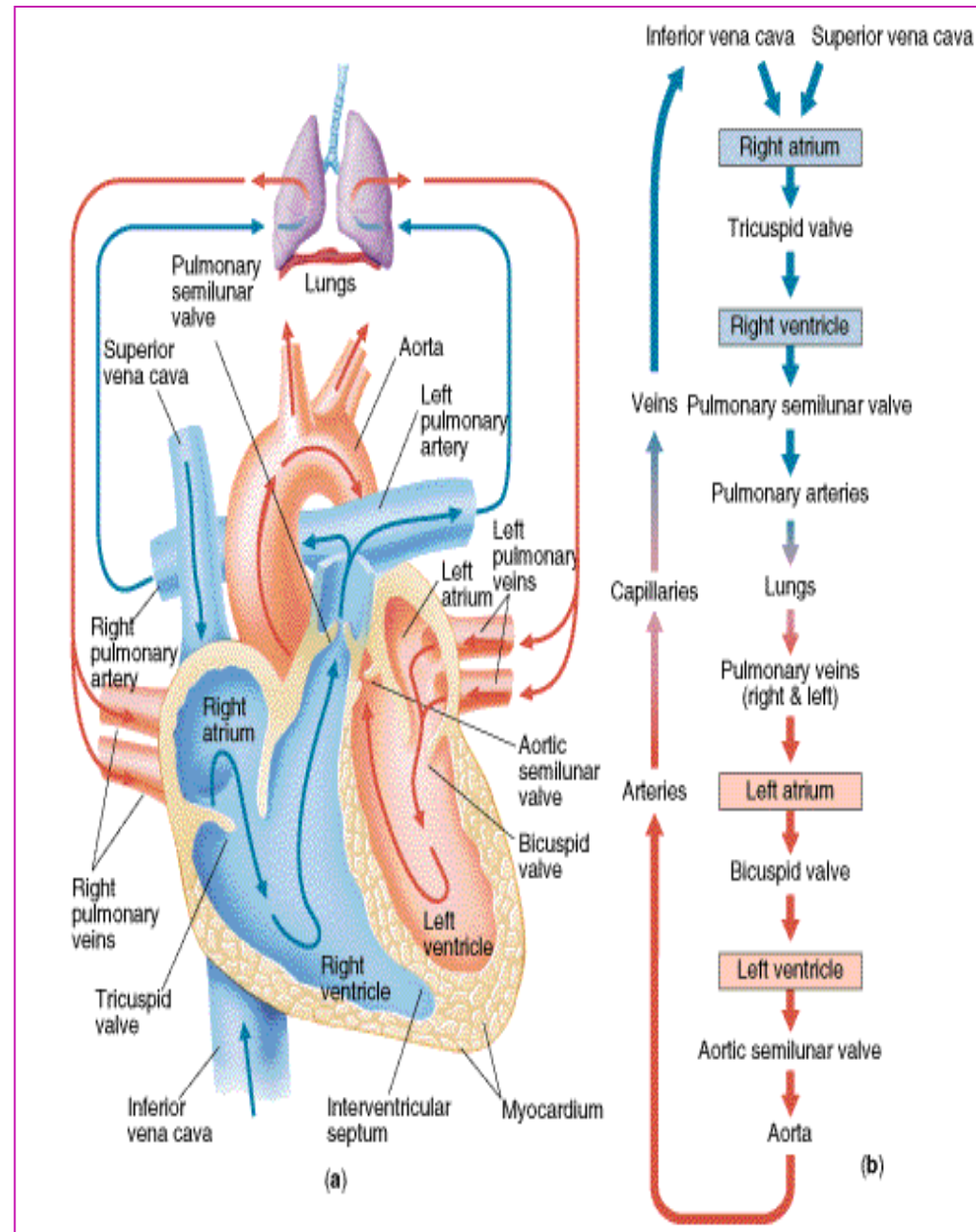


Semilunar valves...

- **SL valves** lack **chordae tendineae** & associated **papillary muscles**.
 - Because of their structural design, they are less likely to experience any form of prolapse.
- All valves consist of connective tissues (not cardiac muscle tissues) and, therefore, open & close passively.
- **Valves** open & close in response to **changes in pressure**:
 - AV valves open when pressure in the atria is $>$ pressure in the ventricles (during ventricular diastole) &
 - AV closed when pressure in the ventricles is $>$ pressure in the atria (during ventricular systole)

Blood Flow through the heart

- Let's begin by examining how blood arrives at the heart.
- **Blood** from the **systemic circulation** contain high **carbon dioxide** and low oxygen.



Blood Flow through the heart

- **Deoxygenated blood** will arrive at the **right atrium** from 3 vessels.

1. Superior vena cava

- drains blood from veins in head, thorax, & upper arms.

2. Inferior vena cava

- drains blood from veins in abdomen, pelvis, & legs.

3. Coronary sinus

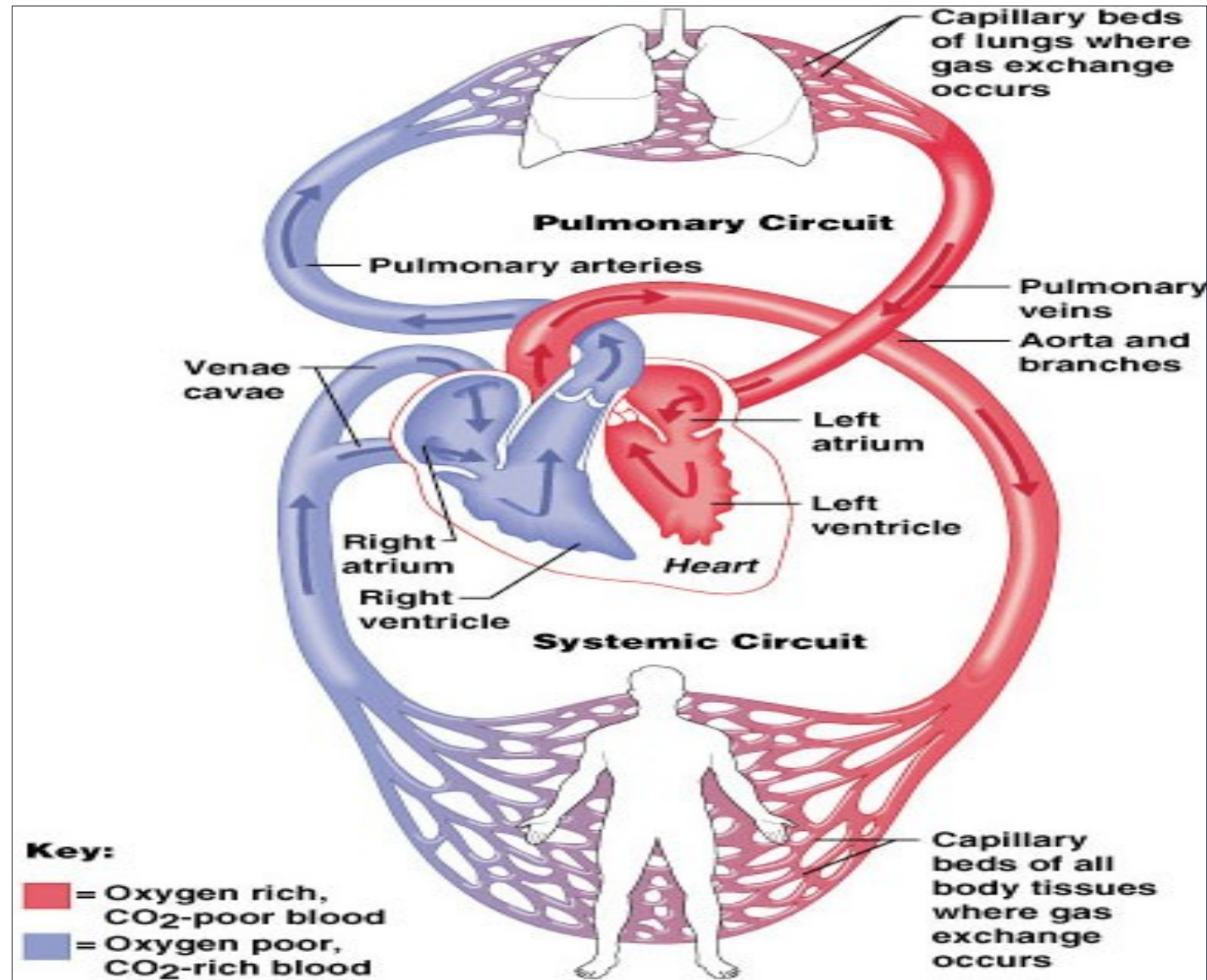
- drains blood from veins in the coronary circulation.

Blood Flow Through the Heart...

- **Blood** from the **pulmonary circulation** is low in CO₂ & high in **O₂** & arrives at the left atrium from 4 blood vessels - the 4 pulmonary veins:
 - right superior, right inferior, left superior, & left inferior pulmonary veins.

- Deoxygenated blood from systemic circulation → right atrium
→ tricuspid valve → right ventricle → pulmonary semilunar valve
→ pulmonary trunk (Oxygenated blood) → pulmonary artery → **lungs**
→ pulmonary veins → left atrium
→ bicuspid valve → left ventricle
→ aortic semilunar valve → aorta → systemic circulation (body).

Blood Flow Through the Heart...



Coronary Circulation

- The heart needs a tremendous amount of **oxygen** & **nutrients** in order to function properly.
- The heart is normally contracting **60-100** times per minute.
- Muscle contraction is an active process and will require energy in the form of **ATP**.
- Cardiac muscle cells are endowed with **lots of mitochondria** to make ATP.

Coronary Circulation...

- Therefore they need a large amount of nutrients (fatty acids, glucose, lactate) and a large supply of oxygen.
 - Because of these needs, the heart has a large blood supply!
- Cardiac muscle is not nourished to any meaningful extent by the blood flowing thru its chambers.
 - So it has its own network of arteries, capillaries, and veins known as CORONARY CIRCULATION.

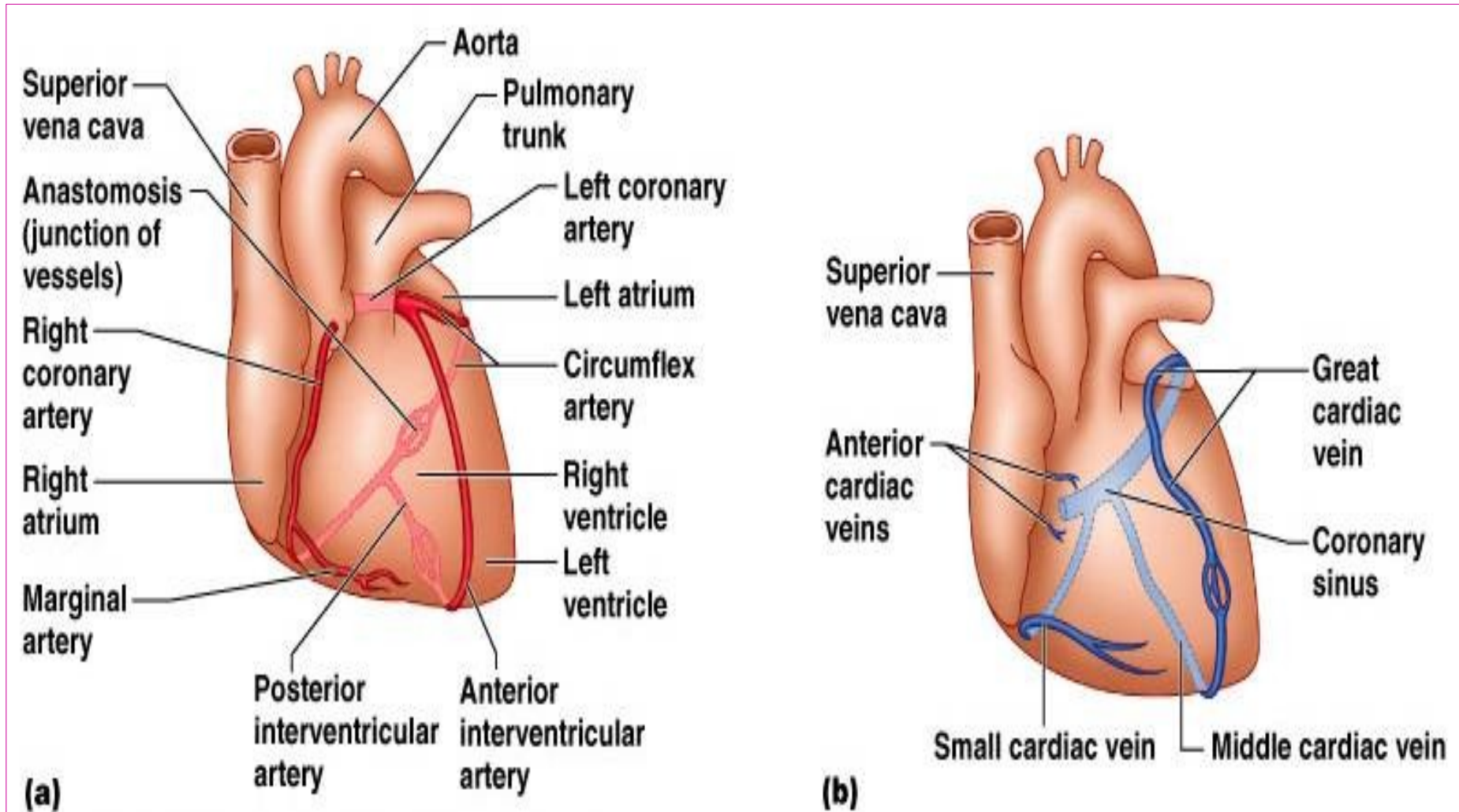
Coronary Circulation...

- **5%** of (cardiac out put) circulating blood is delivered to the heart itself via this coronary circulation.
- After the **aorta** emerges from the left ventricle it gives off 2 branches, the **left** & **right coronary arteries**.
 - The coronary arteries branch throughout the myocardium, spreading out over the heart. Coronary arteries eventually yield **coronary capillaries**.
- Blood then enters the **coronary veins** which combine to produce the **coronary sinus**, which dumps blood into the **right atrium**.

Coronary Circulation...

- The coronary circulation exhibits **collateral circulation** & multiple **anastomoses** (where 2 arteries come together & combine their blood flow).
- Having multiple routes for blood to reach certain areas provides a way to help **ensure the health of cardiac tissue**.
- Consider what would happen if there was 1 blood vessel feeding an area & that blood vessel was **occluded** - those cells that received nutrients from that blood vessel would **die**.
- If there are 2 vessels feeding an area & 1 is blocked, the other can still provide oxygen & nutrients to the cells in the area.

Coronary Circulation...



Coronary Circulation...

- **Ischemia**

- **loss of blood flow** due to blockage of coronary arteries.
- Temporary & reversible ischemia produces a sense of **pain** known as **angina pectoris**.
- Prolonged ischemia (perhaps due to a coronary blockage) can lead to myocardial **cell death** - in other words, a **myocardial infarction** or **heart attack**.

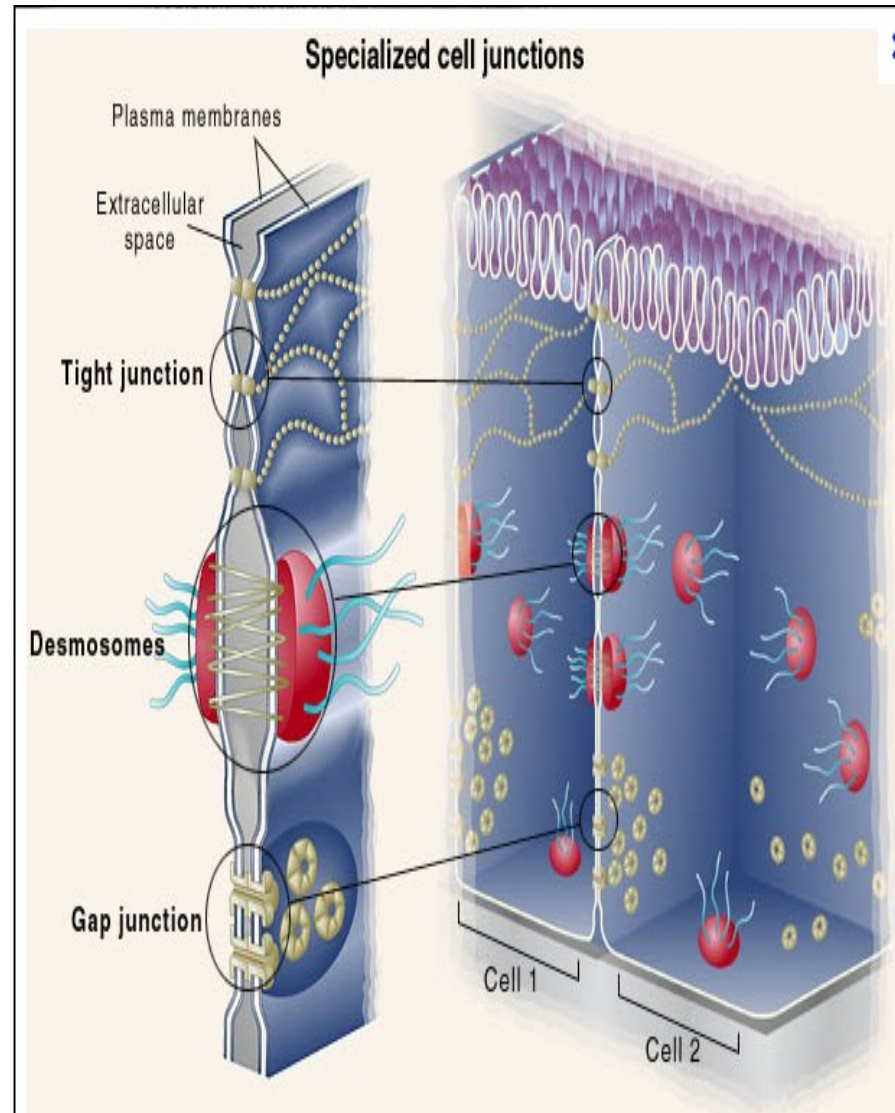
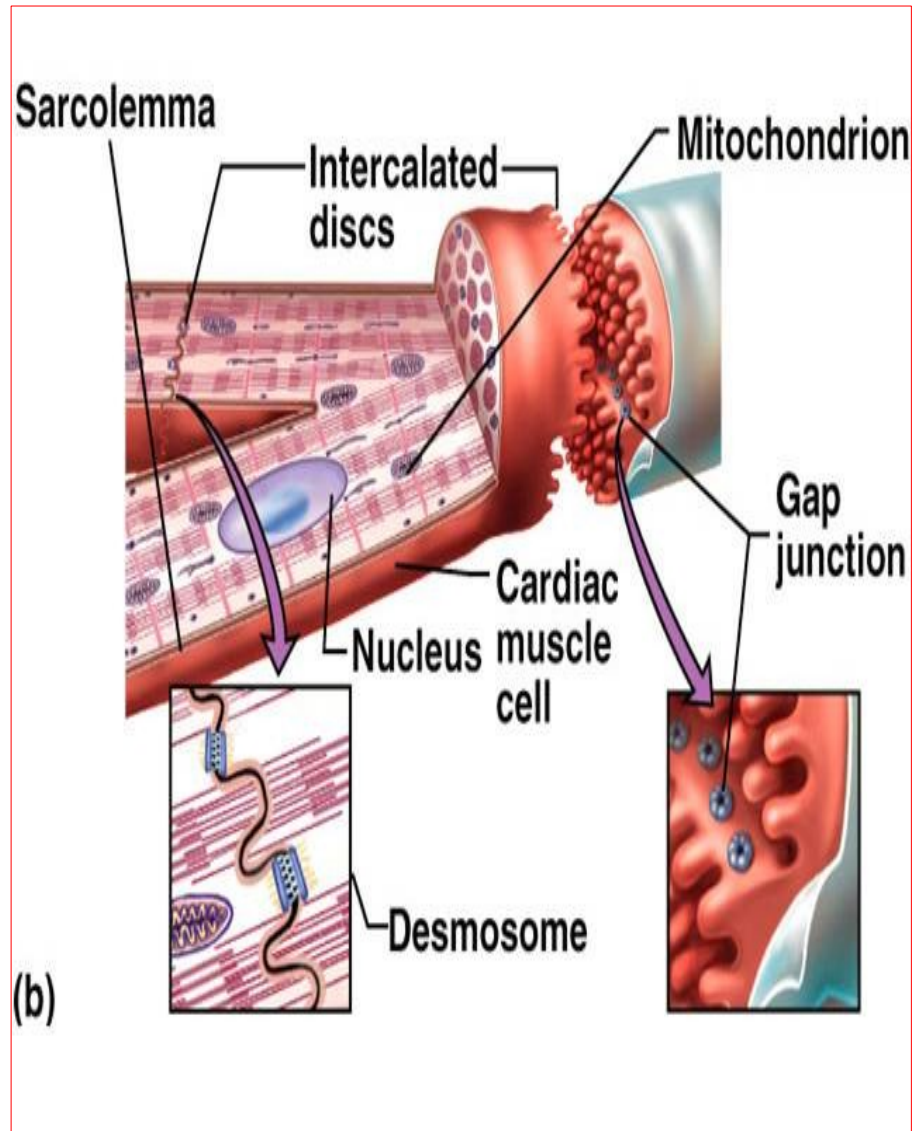
Coronary Circulation...

- In the absence of collateral circulation, the extent of the damaged area during heart attack depends on the size of blocked coronary blood vessel.
 - If larger coronary vessel is blocked, the larger area of heart is deprived of its blood supply.
- Because there are only two major coronary arteries, a blockage of either of these branches results in extensive myocardial damage.
 - Left coronary artery blockage is the most devastating because this vessel is responsible for supplying **85%** of the cardiac tissue.

Cardiac Muscle

- **Myocardium** is the predominant cell type of cardiac muscle cell [**cardiac myocyte**].
- Cardiac muscle cells are **short, thick, branching** and **uni-nucleated**.
 - This is in contrast to skeletal muscle cells which are long, skinny, non-branching and multinucleate.
 - Is similar to skeletal muscle in that they are both **striated**.
- Cardiac muscle cells are intricately linked to one another by structures called **intercalated discs**.

Cardiac Muscle...



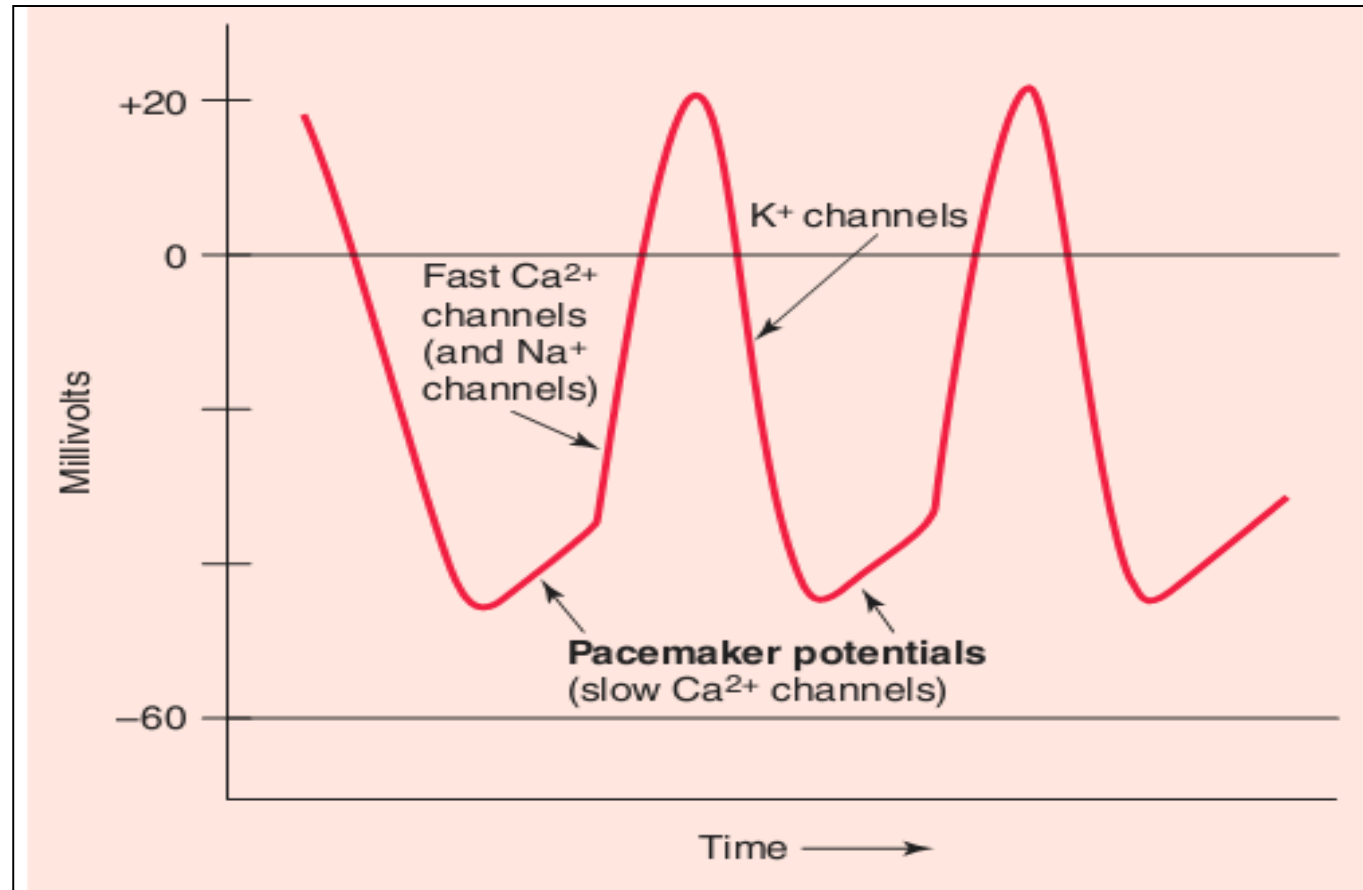
Cardiac Muscle...

- Intercalated discs have 2 components.
 - **gap junctions** (which provide an electrical link between all cardiac myocytes) and
 - **desmosomes** (which provide a mechanical link between all cardiac myocytes).
- The electrical & mechanical connections created by the **intercalated discs** allow the thousands of cardiac muscle cells to behave as if they were one giant cell.
- Multiple cells that function as one entity are often referred to as a **functional syncytium**.

Cardiac Muscle...

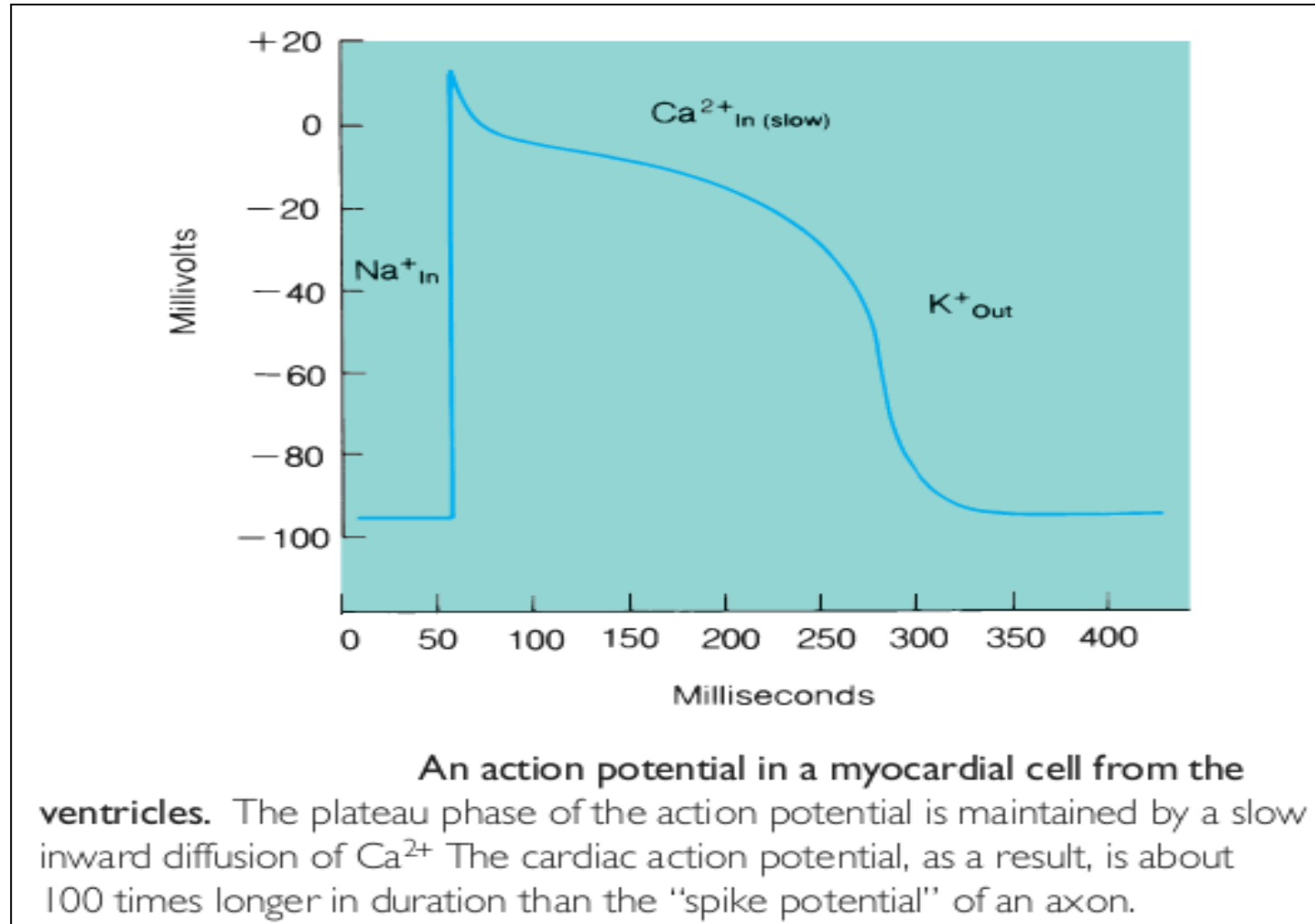
- All cardiac myocytes are not identical.
- **99%** of cardiac wall is composed of the **contractile cardiac muscle cells**.
 - they **generate the force that pumps blood** through the systemic and pulmonary circuits.
- **1%** are the **autorhythmic cells** (pace makers) of the heart.
 - lack the elaborate sarcomeres & other contractile machinery.
 - specialized to **generate action potential spontaneously** and **without nervous system input**.

Pacemaker Potential

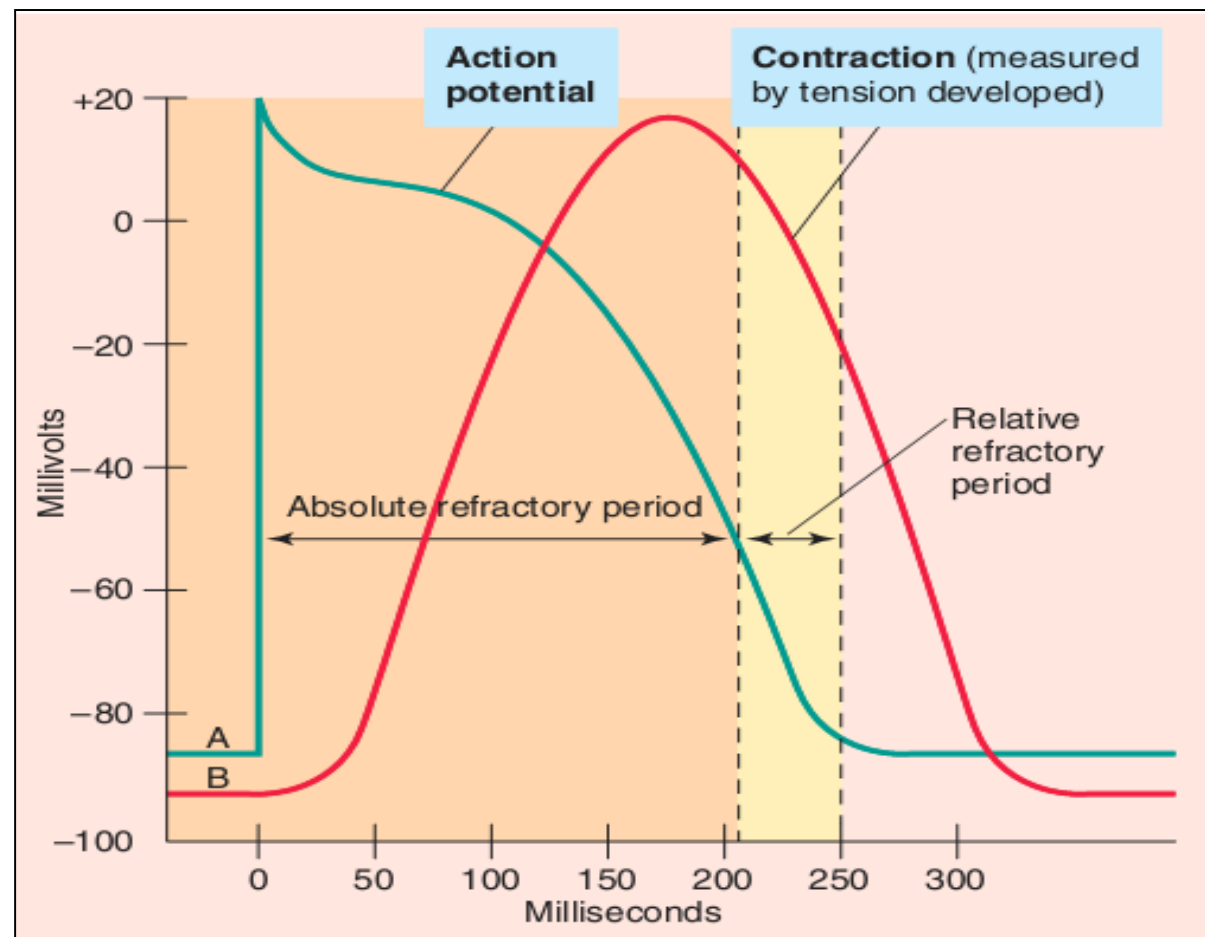


Pacemaker potentials and action potentials in the SA node. The pacemaker potentials are spontaneous depolarizations. When they reach threshold, they trigger action potentials.

Myocardial Action Potential



Myocardial Action Potential...



Correlation of the myocardial action potential with myocardial contraction. The time course for the myocardial action potential (A) is compared with the duration of contraction (B). Notice that the long action potential results in a correspondingly long absolute refractory period and relative refractory period. These refractory periods last almost as long as the contraction, so that the myocardial cells cannot be stimulated a second time until they have completed their contraction from the first stimulus.

Autorhythmic Cells

- "**Autorhythmic**" literally equates to "**self-rhythm**" & that is an appropriate name for these cells because they **set the rhythm of the heart without any input** from any external organs, tissues, or signals.
- They account for the **intrinsic control of the heart rate**, (means by which the heart determines its own rate).
- Autorhythmic cells have the ability to **spontaneously depolarize** to threshold and **generate action potentials**.
- But in case of **skeletal muscle** cells had to be stimulated by **acetylcholine** released by motor neurons in order to generate the action potentials that preceded contraction.

Autorhythmic Cells...

- Masses of autorhythmic cells are found in several locations in the heart:
 1. **Sinoatrial Node** (SA node) - adjacent to the Superior Vena cava opening in the right atrium.
 2. **Atrioventricular Node** (AV node) - near the right AV valve at the bottom of the interatrial septum.
 3. **Atrioventricular Bundle** (AV bundle or Bundle of His) - inferior interatrial septum.
 4. **Bundle Branches** (Left & Right) - interventricular septum.
 5. **Purkinje Fibers** - distributed throughout the right and left ventricle.

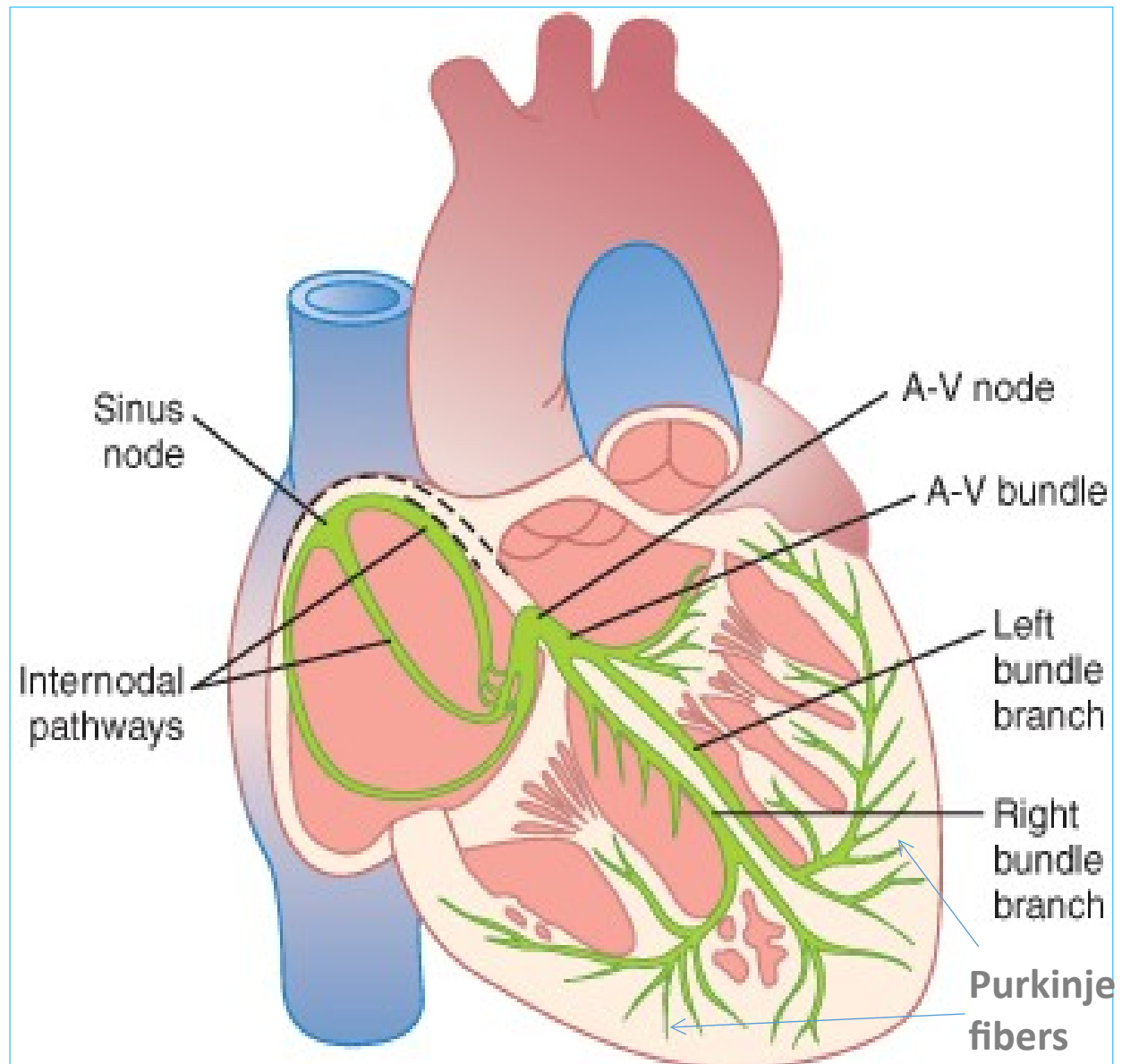
1. **SA node depolarizes.**

2. **Electrical activity goes rapidly to AV node via internodal pathways.**

3. **Depolarization spreads more slowly across atria. Conduction slows through AV node.**

4. **Depolarization moves rapidly through ventricular conducting system to the apex of the heart.**

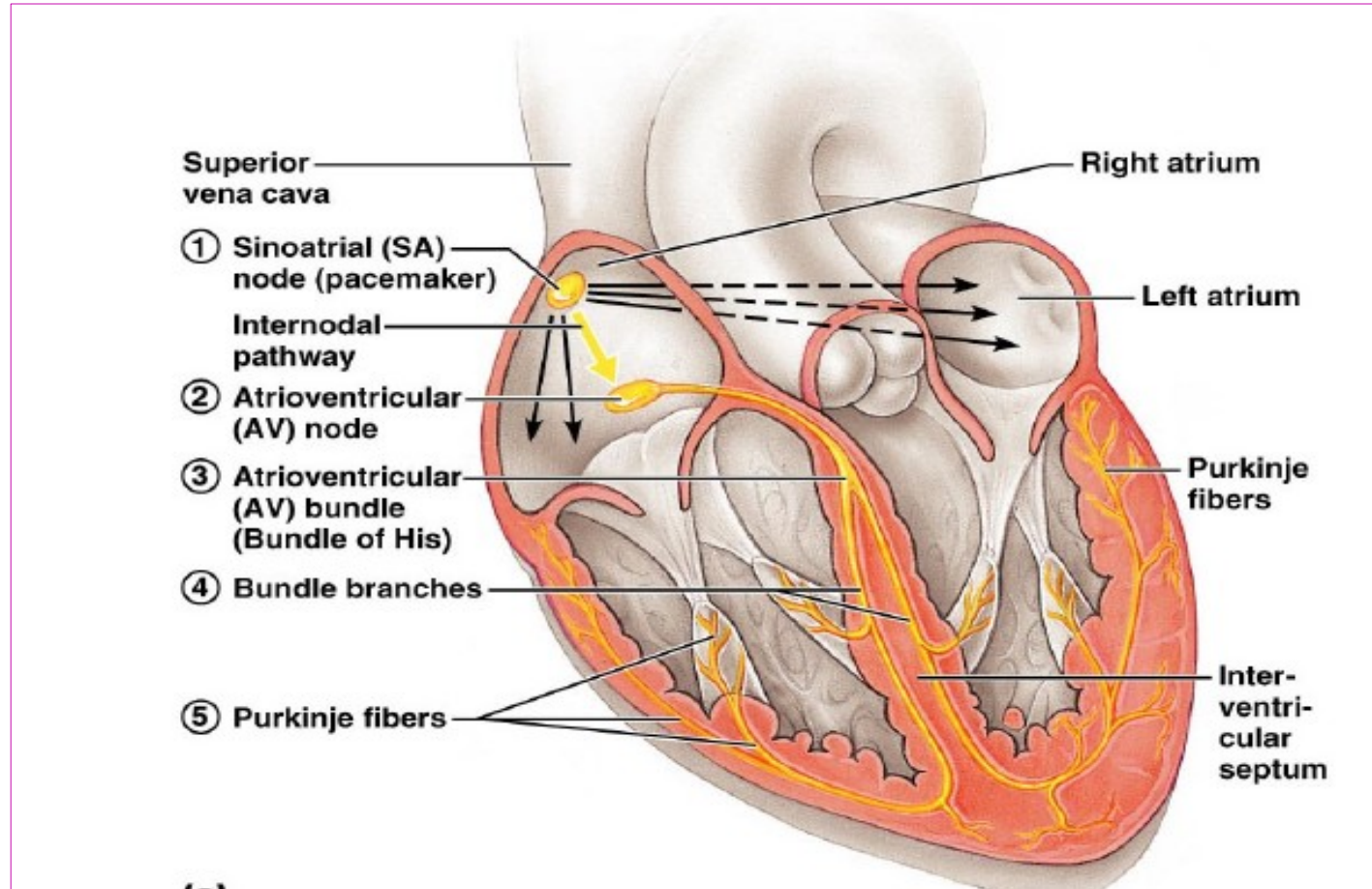
5. **Depolarization wave spreads upward from the apex.**



This diagram shows the route through which the signals from SA node pass through.

Autorhythmic Cells...

SA node -> atrial muscle -> AV node -> bundle of His -> Left & Right Bundle Branches -> purkinje fibers -> Ventricular muscle



Autorhythmic Cells...

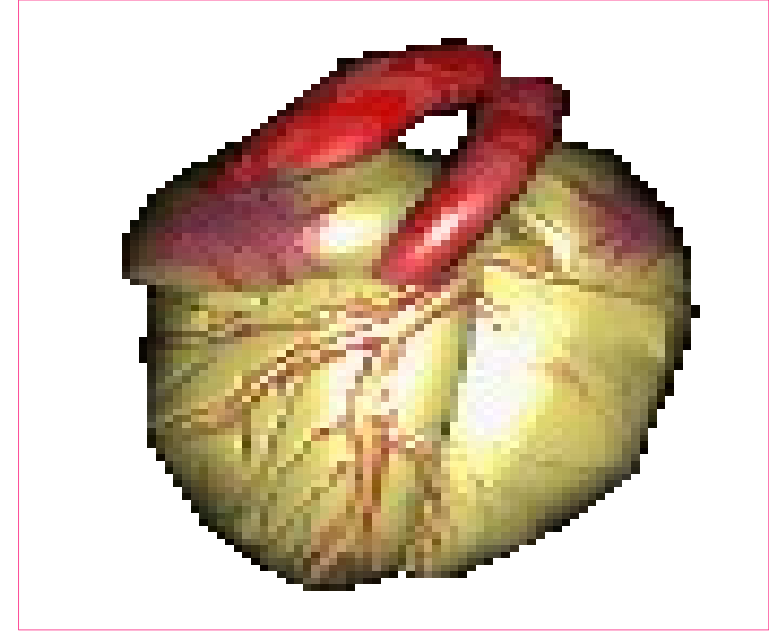
- The autorhythmic cells of the **SA node** have the **fastest rate of depolarization** & thus they will set the heart rate. For this reason, they are known as the **pacemaker of the heart**.
- The **depolarization** that begins in the SA node **travels throughout the atria** to all the **contractile cells** as well as to the cells in the AV node. The atrial contractile cells will respond to the depolarization by contracting.
- At the **AV node**, the **impulse is delayed** momentarily to **allow the atria to complete their contraction** before the ventricles begin to contract.
- From the AV node, the **impulse travels** to the **AV bundle**.

Autorhythmic Cells...

- The **AV bundle** is the only electrical connection between the atria and the ventricles, important for two reasons:
 1. It allows for the **AV delay** which lets the atria complete contraction before the ventricles begin contraction.
 2. It is also involved in making sure the **ventricles begin contracting at the bottom** (apex) rather than the top.
- The impulse travels on to the left & right **bundle branches** & then onward to the **purkinje fibers** which begin at the heart apex & extend upward thru the ventricles.
- By having the **wave of depolarization begin at the bottom**, it allows **contraction to begin at the bottom**.

Autorhythmic Cells...

- Think about why you don't squeeze a toothpaste tube at the top. The same principle applies to the heart.
 - The arteries exiting the ventricles are at the top. Thus to squeeze blood out efficiently, we must begin the contraction at the bottom of the heart!
- From the **purkinje fibers**, the **depolarization travels** to the **ventricular contractile cells** via the **gap junctions** in the intercalated discs.
- The **ventricular contractile cells respond by contracting**. This process then repeats itself over and over and over, with each episode culminating in a **heartbeat**



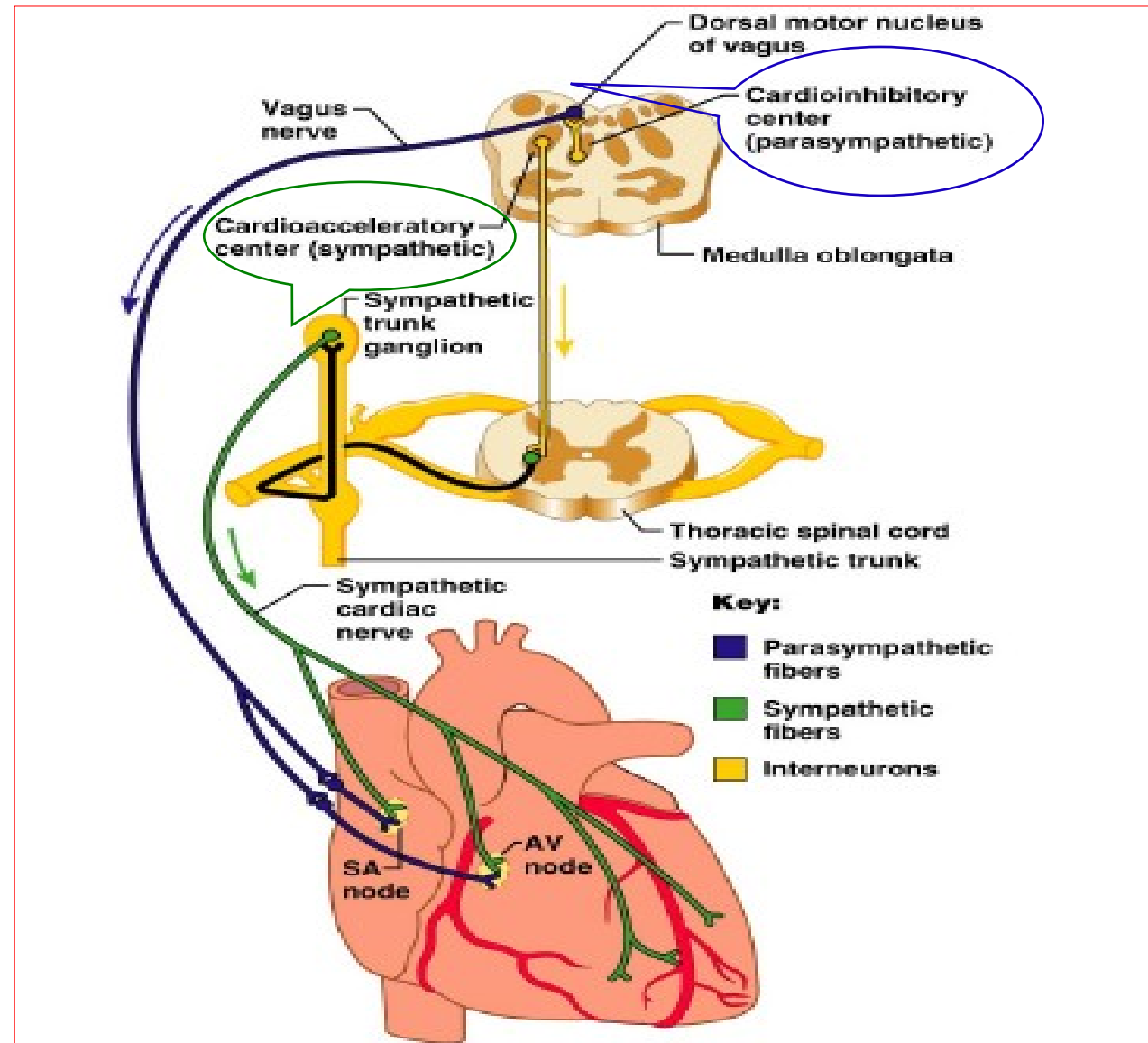
Extrinsic Influences on Heart Rate

- The **autonomic nervous system** provides a large influence on the activity of the heart.
- Increased activity of the **sympathetic nervous system** (the "fight or flight" branch of the ANS)
 - **increases** both the **rate** and the **force** of **heartbeat**.
- Increased activity of the **parasympathetic nervous system** (the "rest and digest" branch of the ANS)
 - **decreases** **heart rate** but has **little effect** on the **force of contraction**.

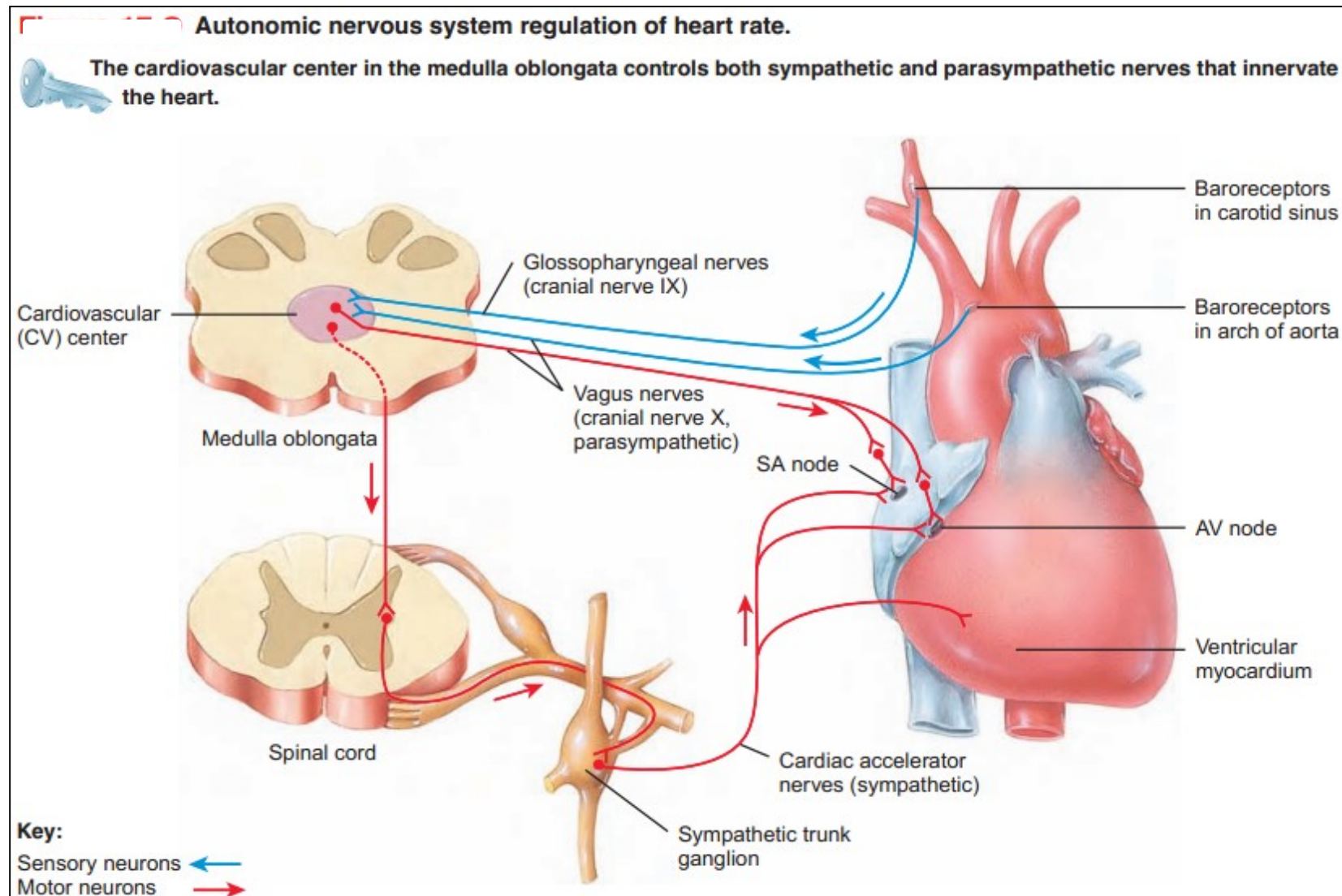
Extrinsic Influences...

- **Medulla oblongata** within the brain stem contains:
 - Cardioacceleratory and
 - Cardioinhibitory center.
- **Cardioacceleratory center** is the source of the **sympathetic output** to the heart.
 - Sympathetic nerve fibers project to the **SA node**, **AV node**, and the bulk of the **myocardium**.
- **Cardioinhibitory center** is the source of the **parasympathetic output** to the heart.
 - Parasympathetic nerve fibers project via cranial nerve 10 (the **vagus nerve**) to the **SA** & **AV** nodes.

Extrinsic Influences...



Extrinsic Influences...



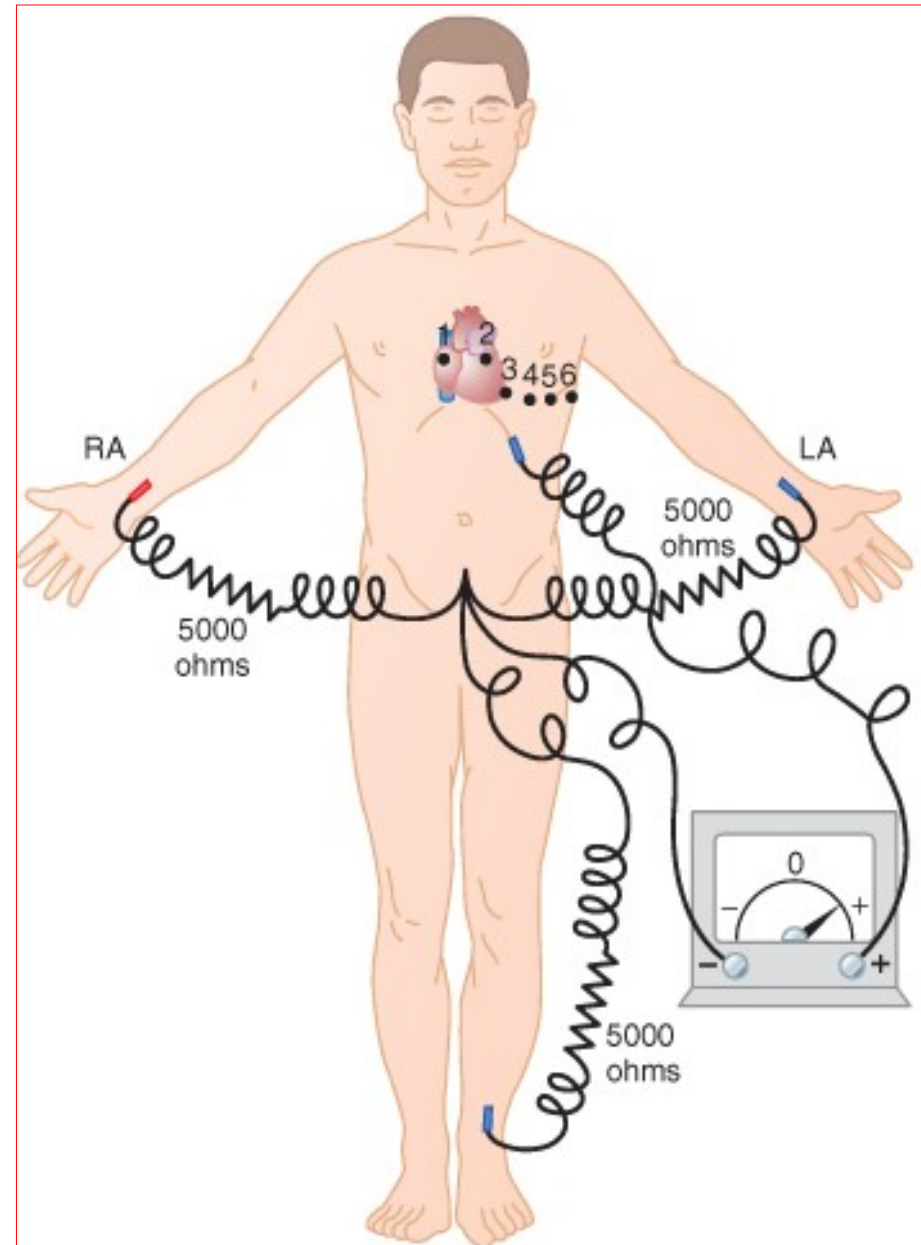
Extrinsic Influences...

- There is a **resting tonic level** of sympathetic and parasympathetic input to the heart.
- **Sympathetic** nerves release **norepinephrine** upon the heart, whereas the **parasympathetic** nerves release **acetylcholine**.
- The **endocrine system** plays a role as various hormones can exert an **effect** on the **heart's rhythm**.

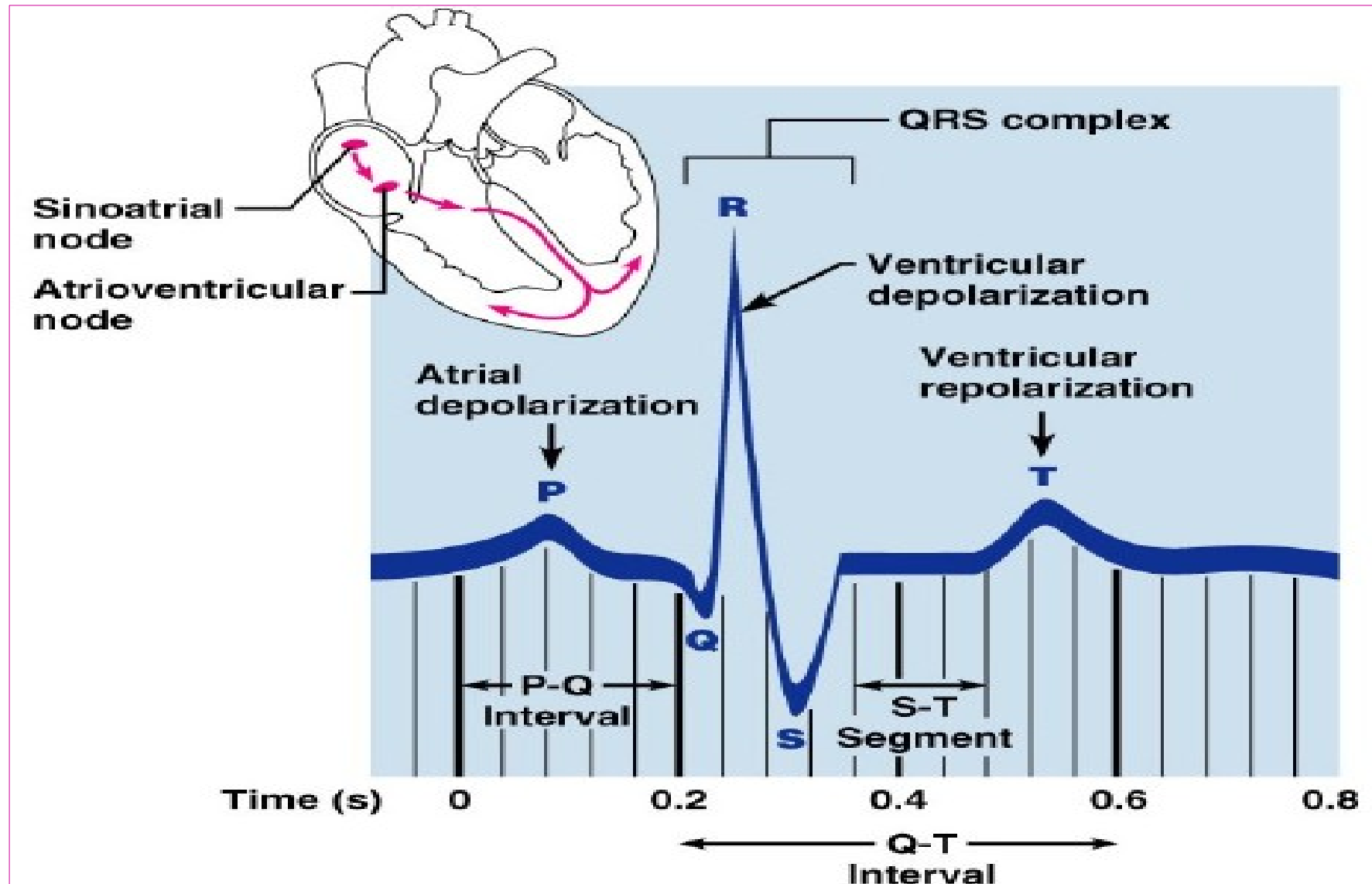
e.g. epinephrine, thyroxine, glucagon

Electrocardiogram [ECG]

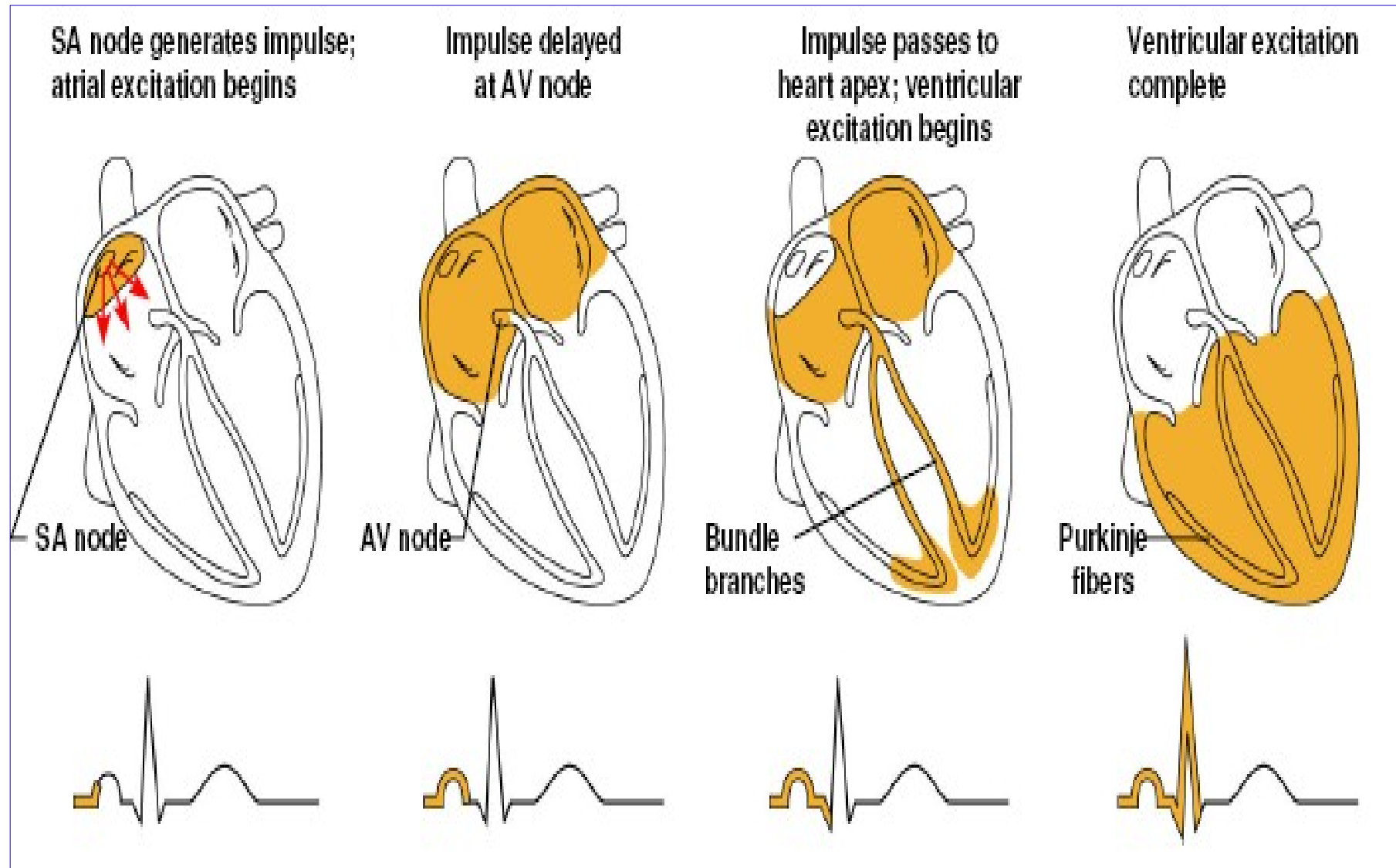
- Record of spread of electrical activity of the heart from the surface of the body.
- The different waves that comprise the ECG represent the sequence of depolarization & repolarization of the atria & ventricles.



ECG...



ECG...



ECG...

❖ P wave

- Represents atrial depolarization (**systole**).

❖ QRS complex

- Represents ventricular depolarization (**systole**).

❖ T wave

- Represents ventricular repolarization (**diastole**).

ECG...

□ PQ interval

- Interval from the beginning of the P wave to the beginning of Q wave . It is the measure of **AV nodal delay**.

□ QT interval

- Interval from the beginning of the Q wave to the end of T wave.
- Represents the entire period of **depolarization** & **repolarization** of the **ventricles**.

□ ST segment

- Segment from the end of S wave to the beginning of the T wave.
- Represents the period within the **entire ventricles are depolarized**.

Function of ECG

- ECG is useful in diagnosing:
 - abnormal heart rates (**arrhythmias**), irregular rhythm of heart rate.
 - **bradycardia**
 - heart rate that is **lower** than 60b/min.
 - **tachycardia**
 - heart rate that is **higher** than 100b/min.
 - conduction abnormalities
 - heart attack
 - coronary artery disease.

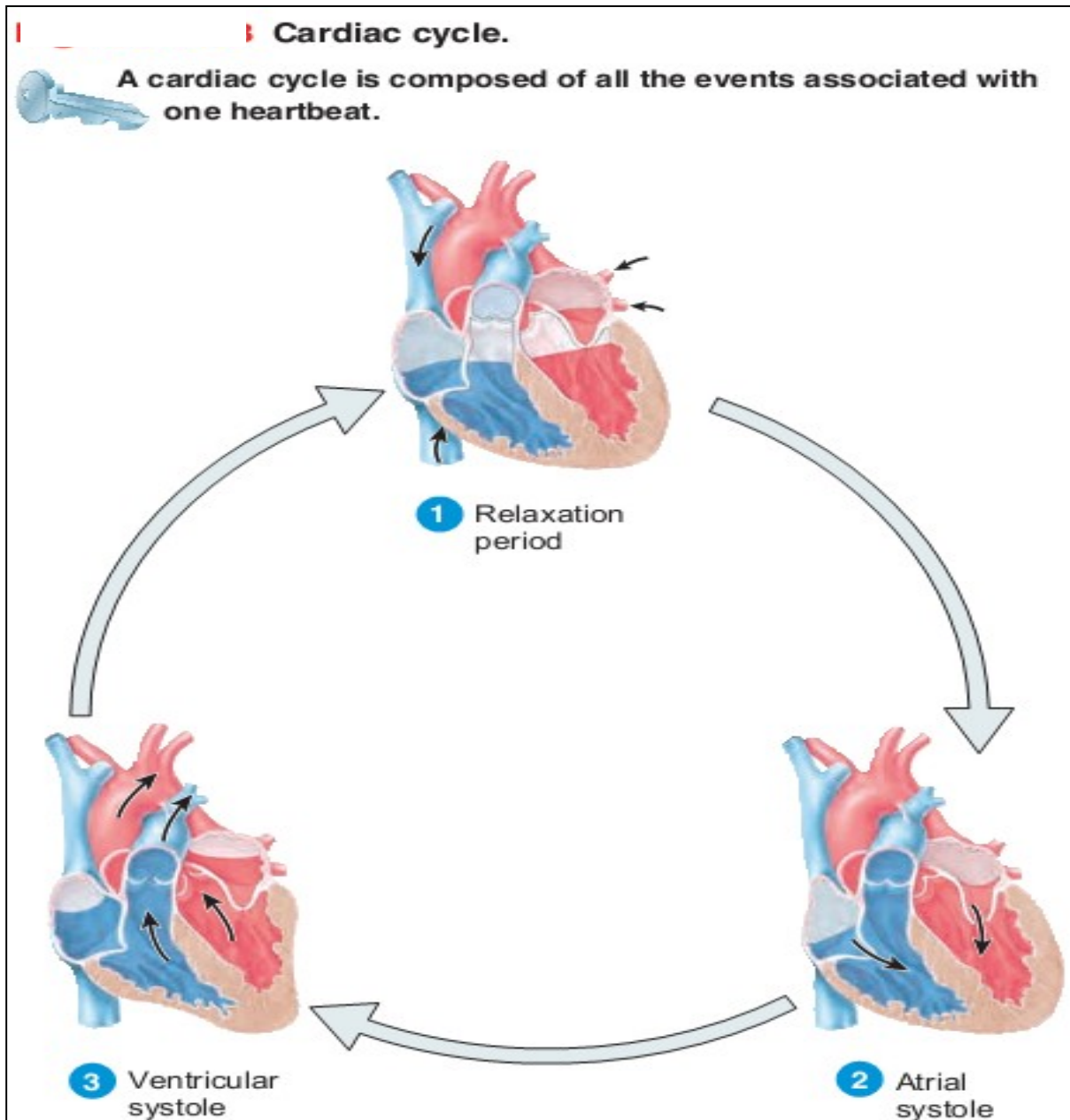
Conduction Problems

Heart Block

- Any **damage** to the **AV node** **interferes** with **transmission of impulses** from atria to the ventricles.
- The damage can be:
 - Ischemia (loss of blood flow due to blockage of coronary arteries).
 - Compression
 - Inflammation
- Varies in severity [1^o block, 2^o block & 3^o block].

The Cardiac Cycle

- **Events** that occur from the beginning of one heartbeat to the beginning of the next.
- Has two major phases:
 - Systole (contraction) & Diastole (relaxation).
- '**Electrical**' events are correlated with '**mechanical**' event.
 - P wave = atrial depolarization = atrial systole.
 - QRS complex
 - = ventricular depolarization = ventricular systole.
 - (atrial diastole occurs at same time)
 - T wave = ventricular repolarization = ventricular diastole.



The Cardiac Cycle...

What happens in the heart during each 'mechanical' event:

I. Atrial systole

- Blood from the atria is pumped into the ventricles.
- Causes a slight **increase in ventricular volume**.

II. Ventricular systole involves:

- A. Isovolumetric contraction
- B. Rapid ejection
- C. Reduced ejection

III. Ventricular diastole

involves:

- A. Isovolumetric relaxation
- B. Rapid filling
- C. Reduced filling
 - diastasis.

IV. Atrial systole

- last rapid filling.

V. Atrial diastole

Ventricular systole

A. Isometric (isovolumic) contraction

- ~ The short period between **AV valve closure** & **opening of SL valve**
 - ventricular contraction doesn't cause blood flow.
- ~ It lasts until the intraventricular pressure exceeds that in arteries [80mmHg in aorta & 10mmHg in pulmonary artery].
- ~ Both **AV & SL valves are in a closed state** → Ventricular pressure rises rapidly (5 to 80 mmHg) but the **volume remains the same**.

Ventricular systole...

B. Rapid ejection

- As soon as the intraventricular pressure exceeds the pressure of aorta, the SL valve opens → rapid ejection of blood through aorta.
- Duration is **0.01sec**, yet **70% volume of blood** is ejected per beat (SV), hence this phase is called so.

C. Reduced ejection

- As the blood is ejected the intravascular pressure decreases from 120 to 80 mmHg & hence the ejection rate declines.
- Duration is **0.08 sec** & **30% of SV is ejected**

Ventricular diastole

A. Isovolumetric relaxation

- The short period between **SL valve closure** & **opening of AV valve**;
 - during which **no** blood flow from atria into ventricles
- There is **no change in ventricular volume** hence called **the period of isometric relaxation**.
-
- It lasts until the intra-atria pressure exceeds that of ventricles.
- The pressure of ventricles falls rapidly as the ventricles continue to relax. Its duration is 0.05 sec.

Ventricular diastole...

B. Rapid filling (period of rapid inflow)

- **ventricular volume increases rapidly**; this occurs because blood that accumulated in the atria during ventricular systole (when the AV valves were closed) now forces open the AV valves (because the pressure in the atria is now greater than the pressure in the ventricles & flows quickly into the ventricles).
- the AV valves are pushed open as a result about **60% blood flows rapidly from the atria to ventricles**. Duration is 0.15sec.

Ventricular diastole...

C. Reduced filling (diastasis)

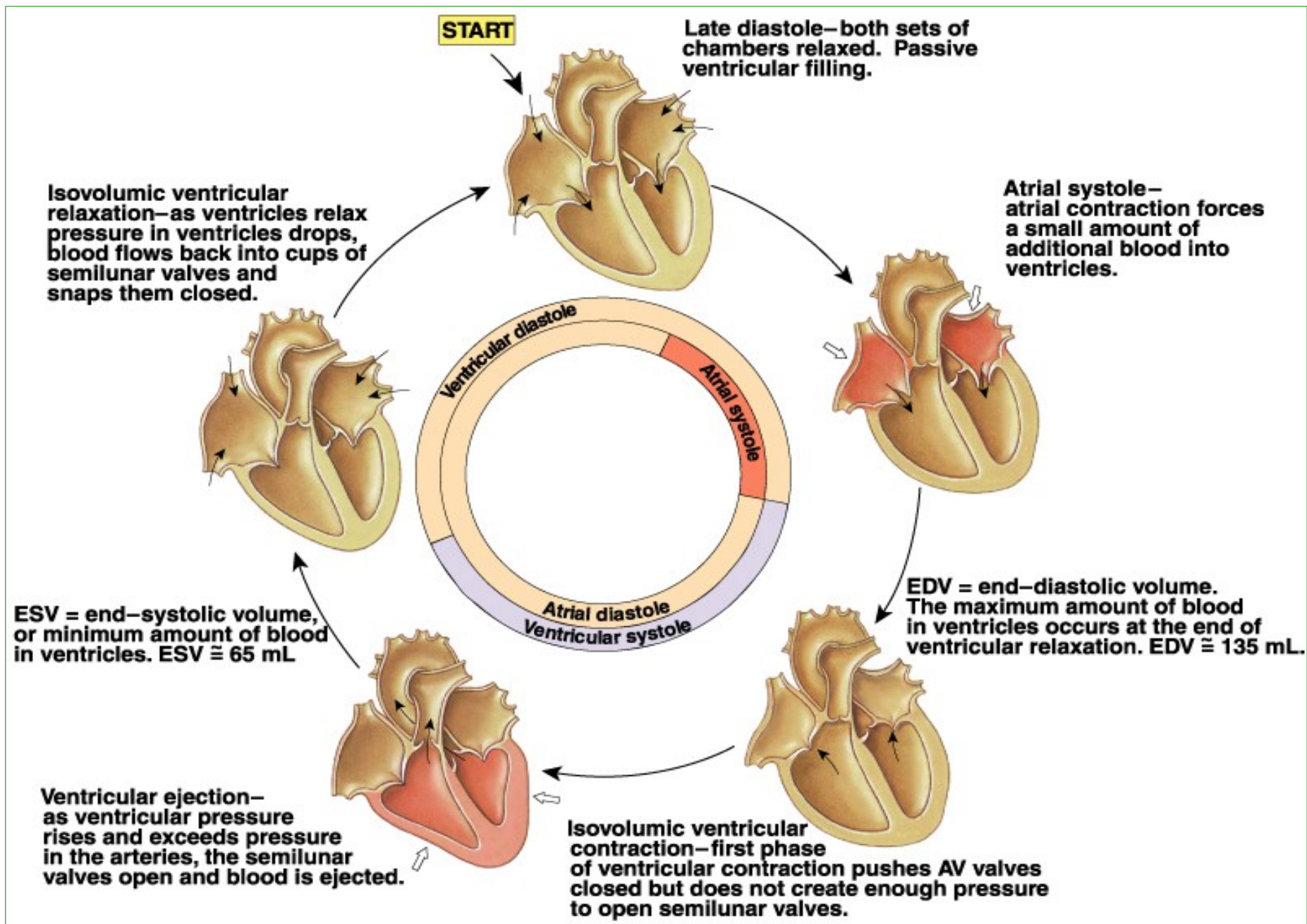
- after the end of rapid filling the ventricular inflow slows down to a considerable extent.
- **10% of Stroke Volume** flows slowly in to the relaxed ventricles

IV. Atrial systole (last rapid filling)

- Atrial contraction which accounts only for about **30%** of ventricular filling.
- It coincides with the 4th heart sound

V. Atrial diastole

- Relaxation of atria, allow filling of atria with blood from veins (**venacava** and **pulmonary vein**).



Heart sounds

Two major heart sounds heard with stethoscope.

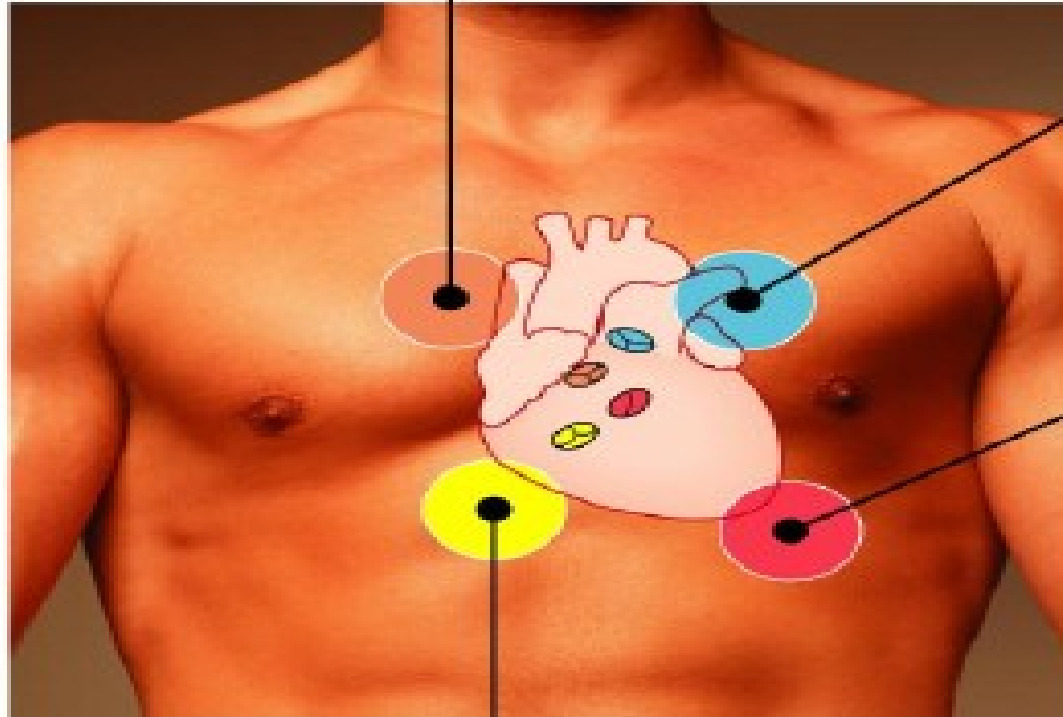
1st heart sound

- Heard as “**lub**” sound.
- Associated with **closure of A-V valve**
- Soft, low-pitched.
- Signals the on set of:
 - **Ventricular systole**

2nd heart sound

- Heard as “**dub**” sound.
- Associated with **closure of S-L valve**
- Sharper, high-pitched.
- Signals the on set of:
 - **Ventricular diastole**

Heart sounds ...



Sounds of aortic valve are heard in 2nd intercostal space at right sternal margin.

Sounds of pulmonary valve are heard in 2nd intercostal space at left sternal margin.

Sounds of mitral valve are heard over heart apex, in 5th intercostal space in line with middle of clavicle.

Sounds of tricuspid valve are typically heard in right sternal margin of 5th intercostal space; variations include over sternum or over left sternal margin in 5th intercostal space.

Heart sounds...

- There exists 3rd & 4th heart sounds.

❑ 3rd heart sound

📖 A weak sound that occurs due to **oscillation of blood** back & forth between **walls of ventricles**.

📖 Coincides with **rapid ventricular filling**.

❑ 4th sound

📖 Is an **atrial heart sound** that occurs during atrial contraction (**atrial systole**).

Abnormal Heart Sounds

- Abnormal heart sounds (**murmurs**) are usually associated with **cardiac diseases**.
- The most common cause of **turbulent blood flow** that produces heart murmurs is **valve malfunction**.
 - **Stenotic** valve & **insufficient** valve
 - Most often, valve stenosis & insufficiency are caused by **Rheumatic fever**,
 - ✓ an autoimmune disease triggered by a streptococcus bacterial infection.

Abnormal Heart Sounds...

1. **Stenotic valve:**

- Stiff/**narrowed valve** that doesn't open completely.
- "**whistling**" sound is heard when such problem exists.

2. **Insufficient valve:**

- the **valves** don't fit together properly, & **can't close completely** due to valve edges scarred.
- the blood flows back ward because of the leaky valves, & thus the opposite direction movement of blood creates a
 - "**swishing**" (gurgling) murmur due to regurgitation of blood

Cardiac Output

- It is a **volume of blood pumped** by **each ventricle per minute**.

$$\text{Cardiac Output} = \text{Stroke Volume} \times \text{Heart Rate}$$

- **Stroke volume**

- The amount of blood pumped out of one ventricle of the heart during each ventricular contraction.
- It is expressed in **ml/beat**.

- **Heart rate**

- The number of beats per minute
- It is expressed in **beats/min**.

Cardiac Output...

- CO is measured in ml/minute & sometimes it is expressed in liters/minute.
- CO varies with the body's state of activity.

Example: CO increases during exercise.

- CO will vary directly with both heart rate and stroke volume.
- Things that can affect HR & SV will thus affect CO.

Cardiac reserve

- The **difference** between:
 - Cardiac output at rest &
 - The maximum volume of blood that the heart is capable of pumping per minute.
- Permits cardiac output to increase dramatically during periods of physical activity.

Factors Affecting Heart Rate

- HR that is persistently **>100bpm** is termed **tachycardia**.
- HR that is persistently **< 60bpm** is termed **bradycardia**.

Increased heart rate can be caused by:

1. ↑ Output of the **cardioacceleratory center**.
~ greater activity of **sympathetic nerves** running to the heart & greater release of **norepinephrine** on the heart.
2. ↓ Output of the **cardioinhibitory center**.
~ less **vagus nerve** activity & a decrease in the release of **acetylcholine** on the heart.

Factors Affecting Stroke Volume

1. Preload

- The amount of **tension** in the **ventricular myocardium** just **prior to contraction**.
- As more blood returns to the heart (↑ venous return in exercise): the more the heart muscle stretches, the greater the force of contraction & the extra blood gets pumped out.

[**whatever comes back to the heart gets pumped out**]

- This direct relationship between **preload** & **stroke volume** is called the **Frank-Starling Law** of the Heart.

Factors Affecting SV...

2. Contractility

- Refers to the **contraction force at** any given **preload**.
- Governed by **neural, hormonal & chemical factors**.
 - ↑ activity of the **cardioacceleratory center** will cause
 - ↑ in frequency of signals along sympathetic nerves to the heart, release of **norepinephrine** onto the heart & **contractility**.
 - The sympathetic hormone **epinephrine** from the **adrenal medulla** also causes **increased contractility**.

Factors Affecting SV...

- The drug **digitalis** is often given to individuals with **congestive heart failure**, causes increased contractility.
- In **myocardial hypoxia**, cardiac myocytes experience a lack of oxygen. With less oxygen, less ATP can be produced, and with less ATP, the muscles cannot generate as much force, and contractility decreases.
- Increased **parasympathetic activity** actually causes **almost no change** in myocardial contractility.
 - this is because there is very **little innervations** of the ventricular myocardium by the vagus nerves.

Factors Affecting SV...

3. Afterload

- Refers to the blood pressure just outside the semilunar valves (**aorta** & **pulmonary trunk**).
- The pressure of blood in these 2 locations opposes the opening of the semilunar valves.
- Remember in order for these valves to be pushed open, the pressure within the ventricles must exceed the pressure within the aorta and pulmonary trunk.

Blood Vessel Types

1. Arteries

- carry blood **away** from the heart.

2. Veins

- carry blood **towards** the heart.

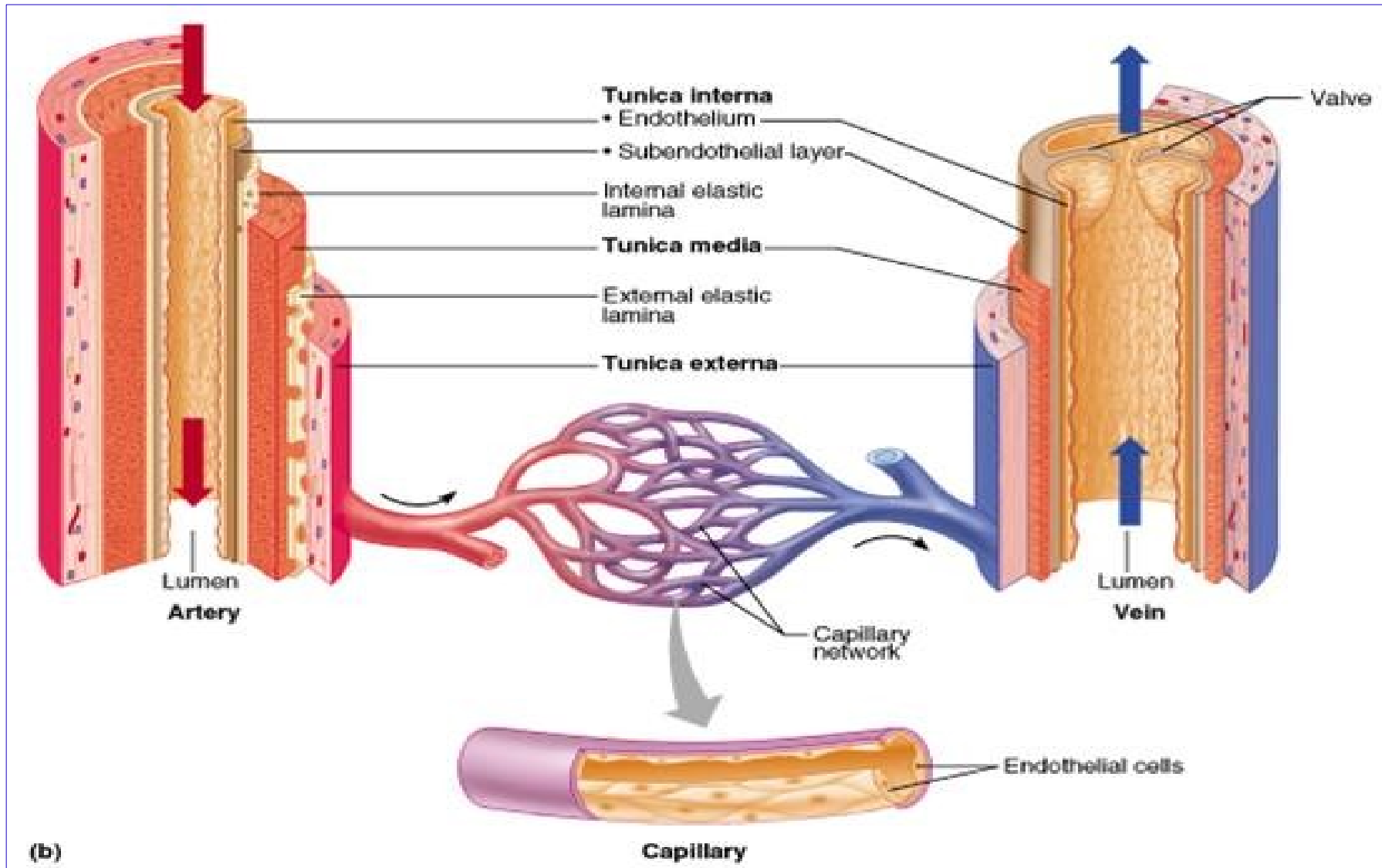
3. Capillaries

- tiny blood vessels that function in the **exchange** of **gases, nutrients, & wastes** between the **blood** & the **interstitial fluid**.

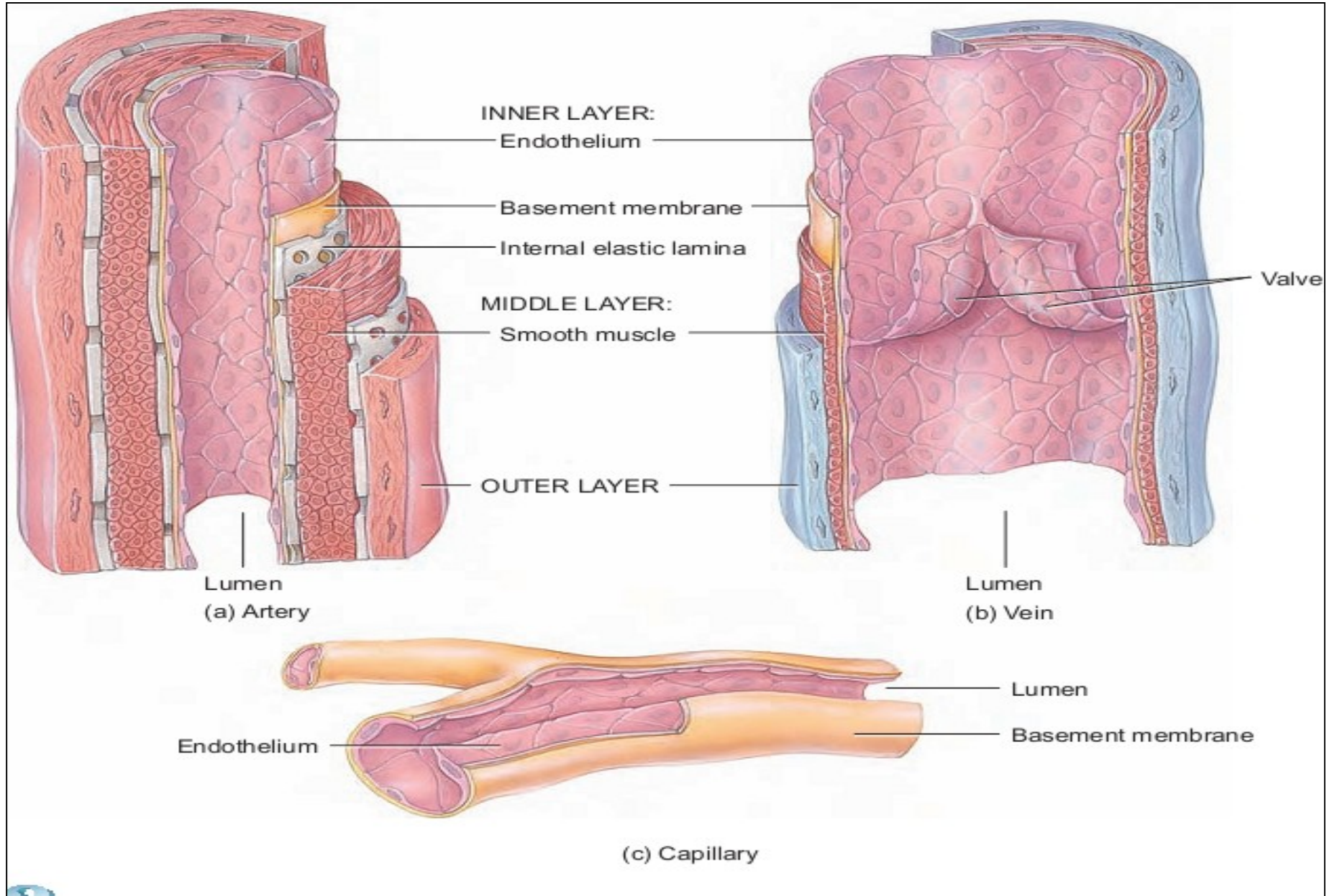
Functions of Blood Vessels

1. Blood vessels form a closed system of tubes that carries blood away from the heart (in arteries), transports it through the tissues of the body (in arterioles, capillaries, and venules), and then returns it to the heart (in veins).
2. Exchange of substances between the blood and body tissue cells occurs as blood flows through the capillaries.
3. Nutrients and oxygen diffuse from the blood through interstitial fluid into tissue cells. Waste products, including carbon dioxide, diffuse from tissue cells through interstitial fluid into the blood.

Blood Vessel Types ...



Blood Vessel Types ...



The walls of arteries & veins have 3 layers that surround the lumen:

1. Tunica externa:

- **Outermost layer.**
- Made primarily of loose connective tissue.
- **Anchors the blood vessel** to the surrounding tissue.

2. Tunica Media :

- Usually the **thickest layer** in arteries.
- Consists primarily of **smooth muscle.**
- Responsible for **vasoconstriction & vasodilation.**

The walls of...

3. Tunica Interna :

- Endothelium overlying sparse connective tissue.
- Acts as a **selectively permeable barrier** to blood solutes.
- Secretes vasoconstrictors & vasodilators.
- Provides a **smooth surface** that repels blood cells & platelets.

Note that capillaries contain **only** a tunica intima.

Arteries

- Constructed to withstand surges of blood pressure associated with **ventricular systole**.
- **More muscular than veins** & appear relatively round in tissue sections.
- There are 3 basic categories of arteries:
 1. Conducting arteries [Elastic arteries].
 2. Distributing arteries [Muscular arteries].
 3. The smallest arteries [Arterioles].

Arteries...

1. Conducting (Elastic) Arteries

- The **largest arteries** including; **aorta, pulmonary arteries** & the common **carotid arteries**.
- Their **tunica media** contains a great deal of **elastic tissue**;
 - allows for **expansion** during ventricular systole & **recoil** during ventricular diastole.
 - maintain blood flow.

Arteries...

2. Distributing (Muscular) Arteries

- Smaller **branches** that distribute blood to individual organs.
- They typically have 25-40 layers of **smooth muscle** cells.
- Examples include the **brachial**, **femoral**, and **splenic arteries** .

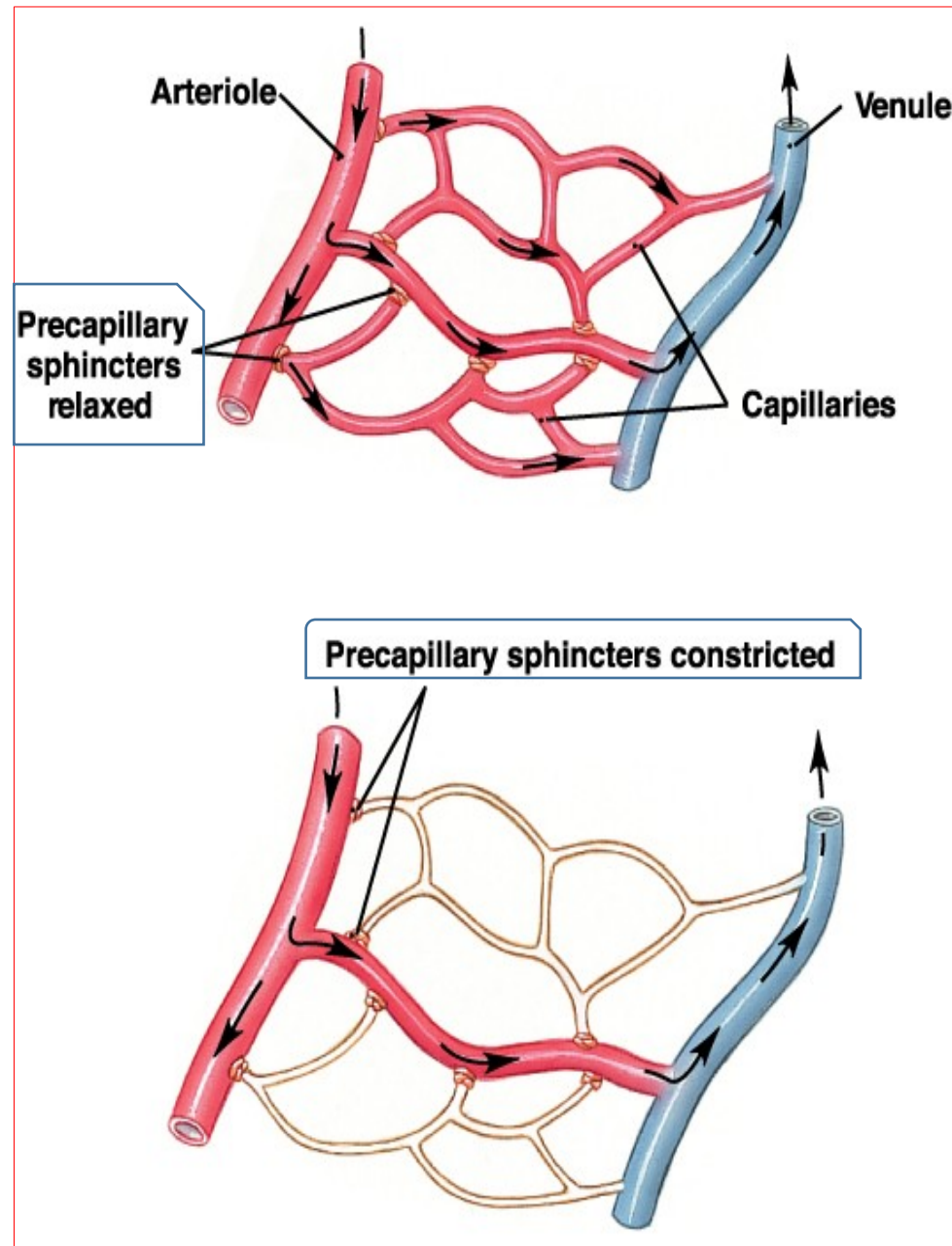
Arteries...

3. Arterioles (the smallest arteries)

- **Heavily innervated** & the 1^{ry} points at which the body controls the relative amounts of blood directed to specific organs.
- Linking the arterioles to the capillary beds are short vessels known as **metarterioles**.
- Instead of a continuous tunica media, they have **individual muscle cells** that are spaced apart.

Arteries...

- These muscle cells form **precapillary sphincters** which encircle the entrance to a capillary bed.
- These sphincters **regulate** how much **blood** will **flow** through particular capillary beds.



Capillaries

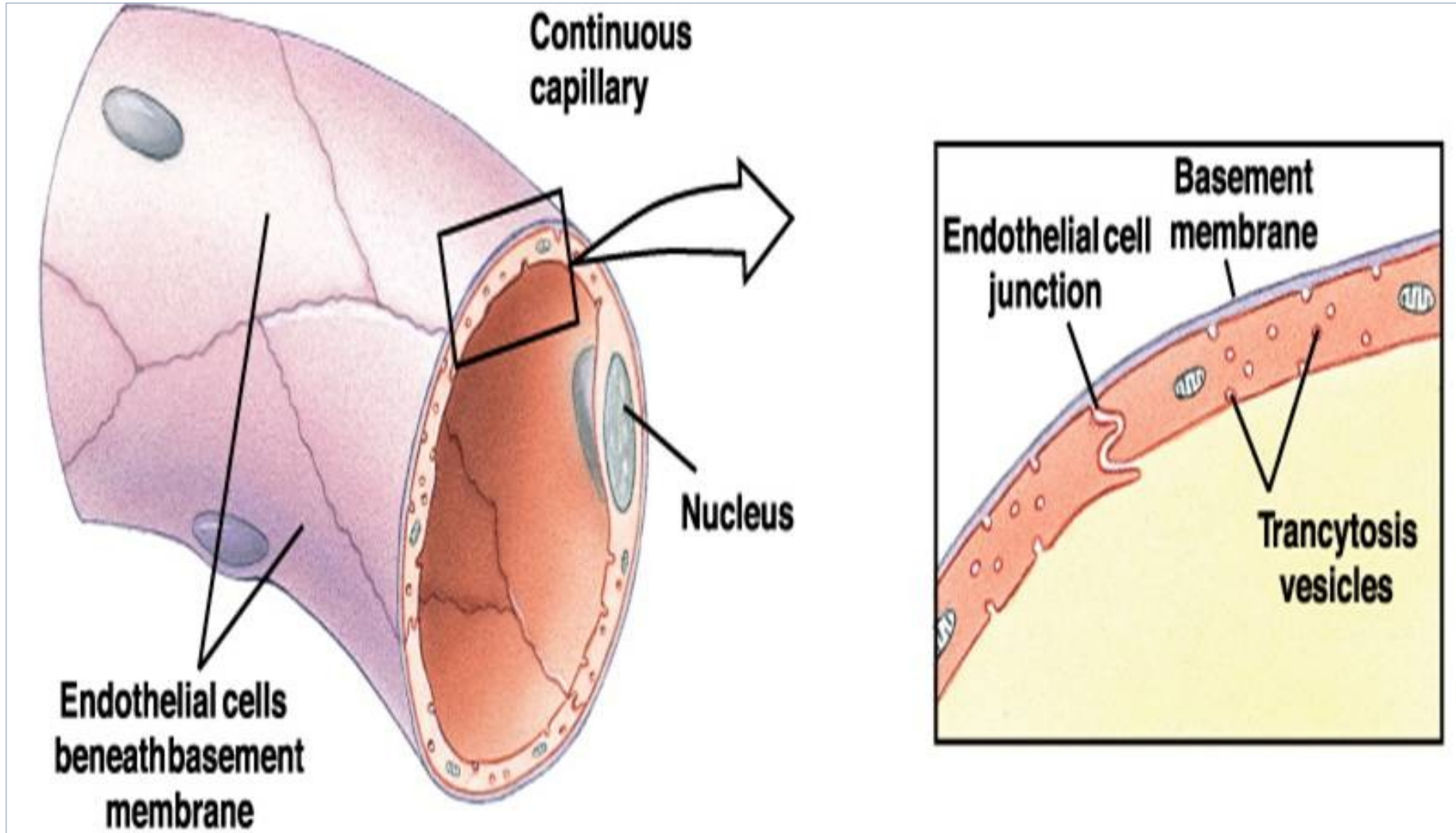
- Area where **exchange** occurs, contain only an endothelium & a basement membrane.
 - Approximately **1billion** in the human body.
 - Lots of capillaries means **lots of surface area**, the more surface area, the **easier it is to exchange** **gases, nutrients, wastes**, etc.
 - Organized into groups of 10-100 in capillary beds.
- Of 3 types: continuous, fenestrated & sinusoidal.

Capillaries...

1. Continuous Capillaries

- Are most common, abundant in **skin** and **muscle**.
- Endothelial cells are joined by **tight junctions**, **but** also contain **intercellular clefts** through which small molecules (e.g. glucose, but not albumin) can pass.
- Note that cerebral capillaries **lack these clefts** and have far more numerous tight junctions.
 - helps protect the delicate brain tissue from blood-borne toxins and pathogens.

Capillaries...

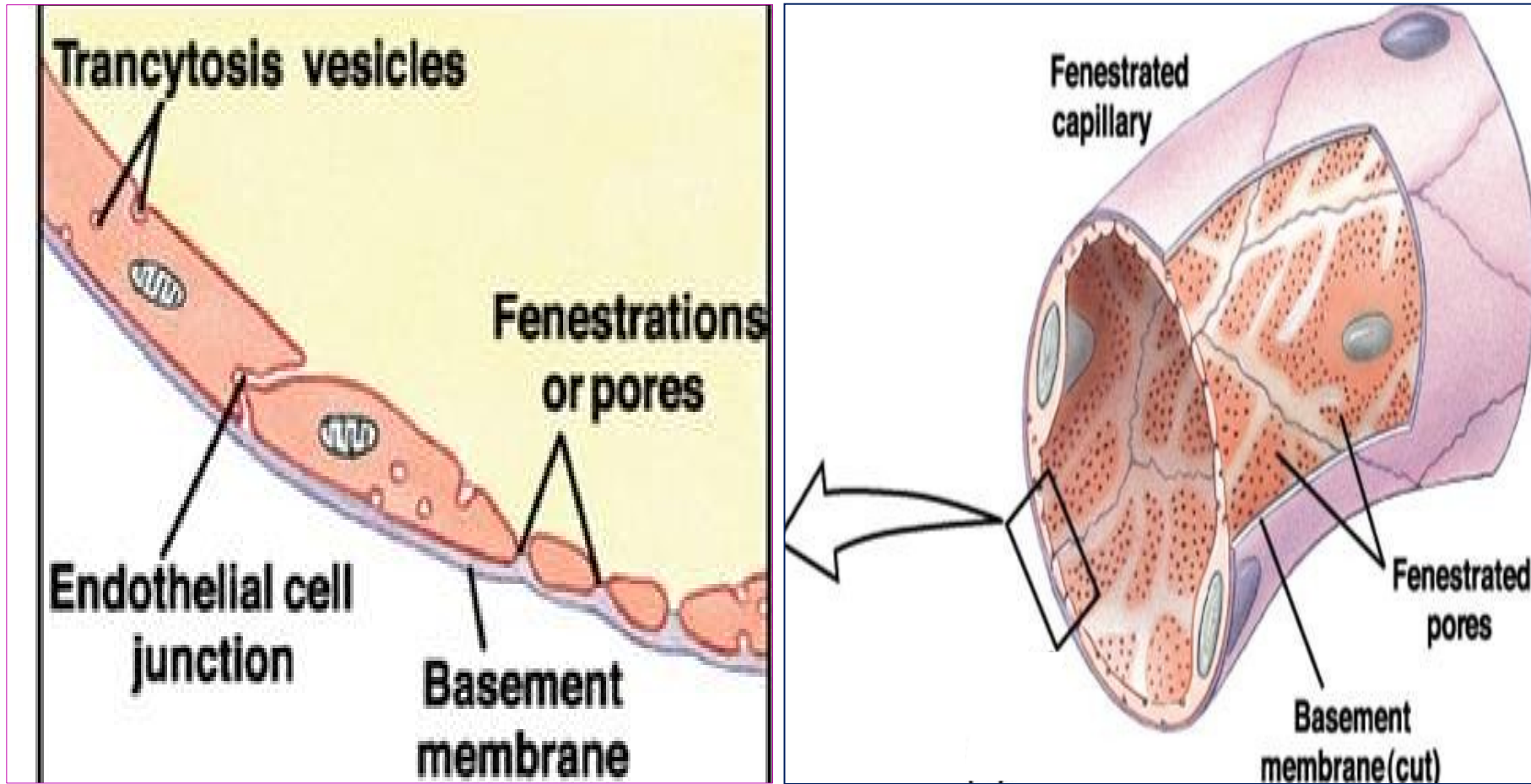


Capillaries...

2. Fenestrated Capillaries

- Similar to continuous capillaries but some of the endothelial cells are riddled with pores-fenestrations.
- Found in sites of :
 - active absorption (small intestine),
 - secretion (endocrine organs) and
 - capillary filtration (kidneys).

Capillaries...

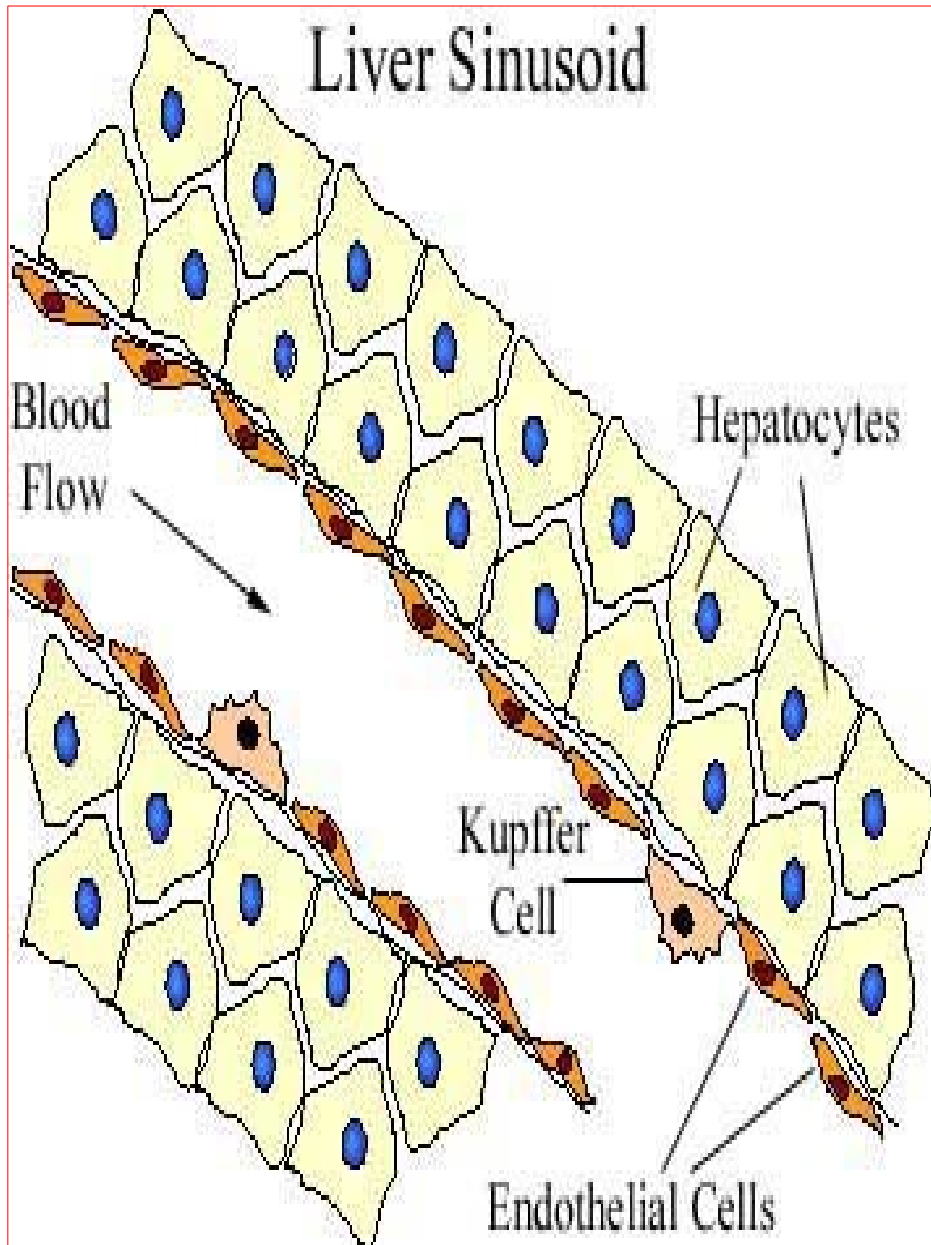


Capillaries...

3. Sinusoidal Capillaries

- Highly modified, **extremely leaky**, fenestrated capillaries.
- Contain irregularly shaped lumen & **large intercellular clefts**.
- Found in areas where **large stuff** needs to **exit** or **enter** the bloodstream such as:
 - **bone marrow** (passage of nascent blood cells).
 - **lymphoid organs** (for easy entry/exit by WBCs).
 - **liver** (for large plasma proteins, e.g., albumin).

Capillaries...



- In the **liver** & **spleen** the endothelium is intimately associated with macrophages.
- In these locations the sinusoids are **twisty & tortuous**.
 - makes **blood flow extra slowly** which **gives time** for splenic and hepatic **macrophages** to **monitor** and **assess** its contents.

Veins

- Blood flows from the **capillaries** into **venules** and venules join to form **veins**.
- Veins have **3 tunics**, but the **walls are thinner** & the **lumens larger** compared to arteries.
- In veins the **tunica media** is usually somewhat **small** and the adventitia is the dominant layer.
- Because they are **so far from the heart**, **pressure** in the veins is typically **quite low** (10mmHg) and does not fluctuate with the heartbeat.

Veins...

- Veins are **quite distensible** & their **thin walls** allow them to act as **reservoirs of blood** (approximately **65%** of the body's blood supply is found within the veins).

- Considering the relatively low pressure in the veins, how blood is forced through them to get back to the heart ?

- Since there is a minimal pressure gradient, combination of 3 different factors provide the **driving force**. These are:
 - Respiratory & skeletal muscle pump, valves.

Veins...

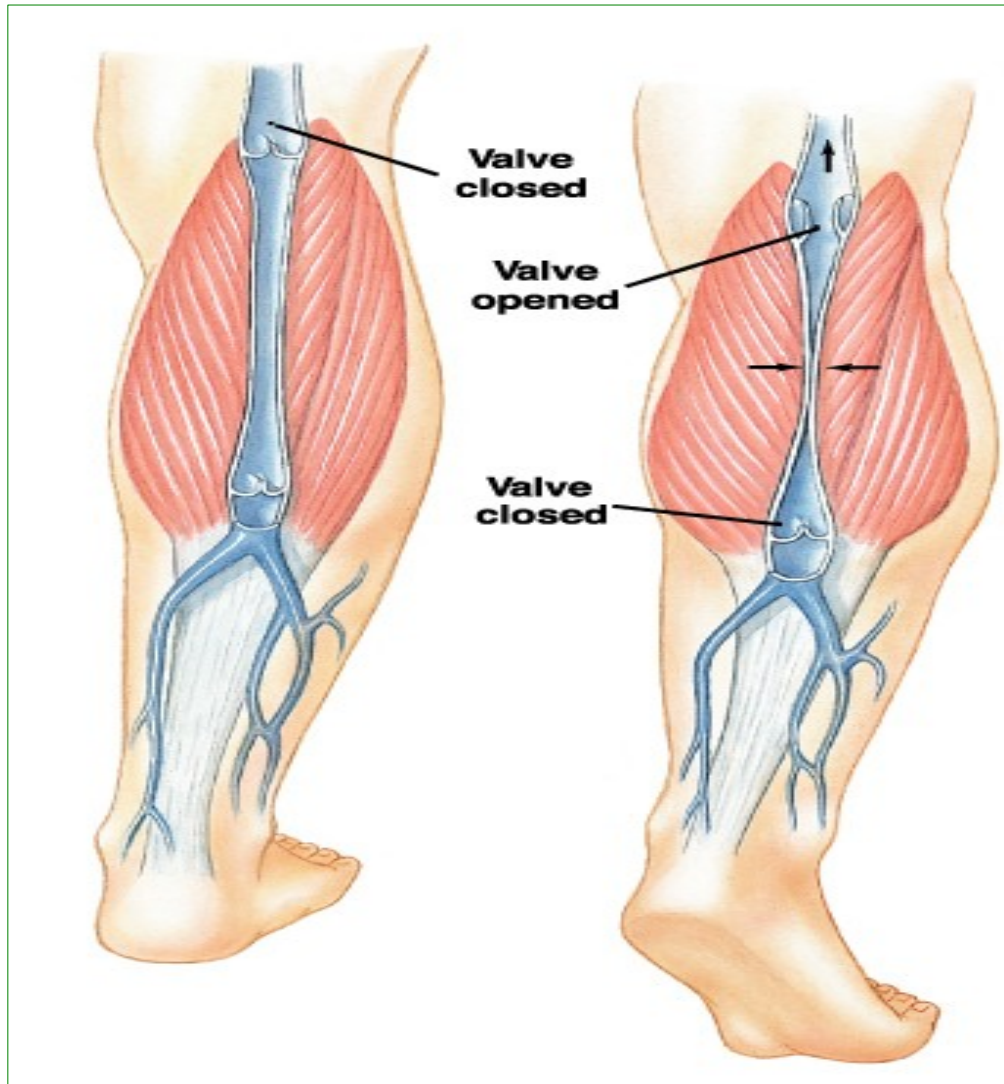
1. Respiratory Pump

- As we inhale, our thoracic cavity expands while our **abdominal cavity compresses**.
- This **increase pressure in the abdominal veins** resulting in an **increased blood flow** towards the heart.

2. Venous Valves

- One way valves made of flaps of endothelium are found in medium veins (mostly in the **legs** & the **arms**) where they help **prevent backflow**.

Veins...



3. Skeletal Muscle Pump

- the contraction or relaxation cycles of skeletal muscles **squeeze the veins** forcing the contained blood towards the heart.

Veins...

- In people who stand for long periods of time, blood tends to pool in the lower limbs and **stretch the veins**.
- This is primarily because the lack of movement means a **lack of skeletal muscle pump** activity.
- **Stretching** can **pull the cusps of the venous valves** farther and farther apart until the **valves are unable to prevent backflow**.

Veins...

- The **veins** can then become even further **distended** & **twisted** and then they are known as **varicose veins**.
- In **obese** & **pregnant** individuals, the **pelvic veins** are **compressed**. This retards the return of blood from the legs.
 - With more blood remaining within the leg veins, the likelihood of varicose veins in such individuals increases.
- Varicose veins of the anal canal are known as **hemorrhoids**.

Control of blood vessel diameter

Vasoconstriction

- A decrease in blood vessel diameter.
- Caused by **contraction** of the **smooth muscle** in the vessel's tunica media.

Vasodilatation

- An increase in blood vessel diameter.
- Caused by **relaxation** of the **smooth muscle** in the vessel's tunica media.

Control of blood vessel ...

- Most changes in blood vessel diameter occur in the **arterioles** which contain a considerable amount of **sympathetic innervations** but only sparse parasympathetic innervations.
- There is usually a **tonic release** (a constant resting release) of **norepinephrine** on arteriolar smooth muscle. This release of **NE** can be:
 - **Increased** to cause **vasoconstriction** or can be
 - **Decreased** to cause **vasodilation**.

Blood Flow

- The **amount of blood flowing** through an **organ**, a **tissue**, or a **vessel** in a given time (mL /min).
- If blood flow does not keep pace with tissue metabolism, then **tissue necrosis** (death) can occur.
- In a resting individual total flow is quite constant and equal to **cardiac output**.
- **Flow** through individual organs **varies** from minute to minute as blood is redirected from one organ to another.

Blood Flow ...

- Blood flow between point A & B depends primarily on the **pressure gradient** between the 2 points & the inherent **resistance** of the vessel & the fluid.
- Flow is directly proportional to the pressure gradient & inversely proportional to the resistance.

$$\text{Flow Rate (F)} = \Delta P / R$$

Blood Pressure

- **Hydraulic force** that **blood exerts** upon the **vessel walls**.
- It is determined primarily by:
 1. Cardiac output
 2. Total peripheral resistance and
 3. Blood volume.
- **Resistance**
 - It is **hindrance to blood flow** through a vessel caused by friction between blood & vessel walls.

$$R = 8\eta l / \pi r^4$$

Blood Pressure ...

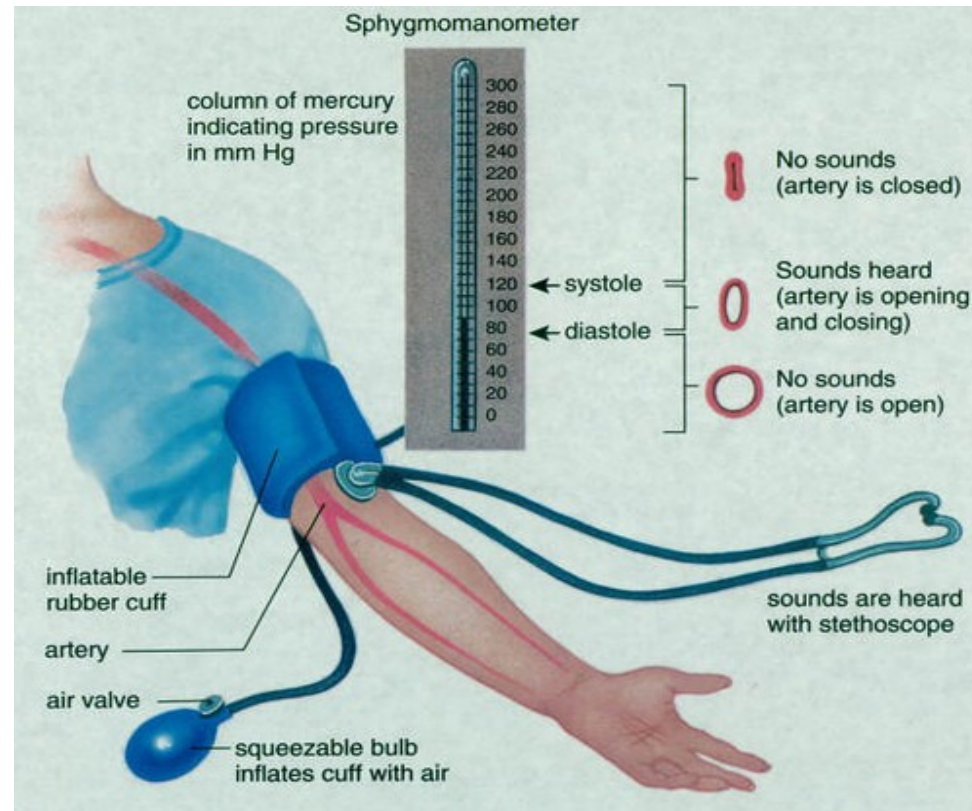
- As the **left ventricle contracts**, blood is ejected into the arterial system.
 - This forces the blood already within the arteries onward.
- However, more blood enters the arteries than leaves.
 - Thus with each heartbeat, we see a rise in the pressure exerted by blood onto the blood vessel walls.
- The **peak pressure** is known as the **systolic pressure** & its normal value is **120mmHg**.

Blood Pressure...

- As the **heart relaxes**, the elastic walls of the arteries rebound and blood leaves, entering the capillaries.
 - As blood leaves, the pressure drops and drops and drops until the next ventricular contraction.
- The **lowest pressure** that is achieved between beats is known as the **diastolic pressure** & its normal value is **80mmHg**.

Blood Pressure...

- Arterial pressures are commonly measured with a **stethoscope** and a **sphygmomanometer**.



Arterial blood pressure

❑ Systolic pressure

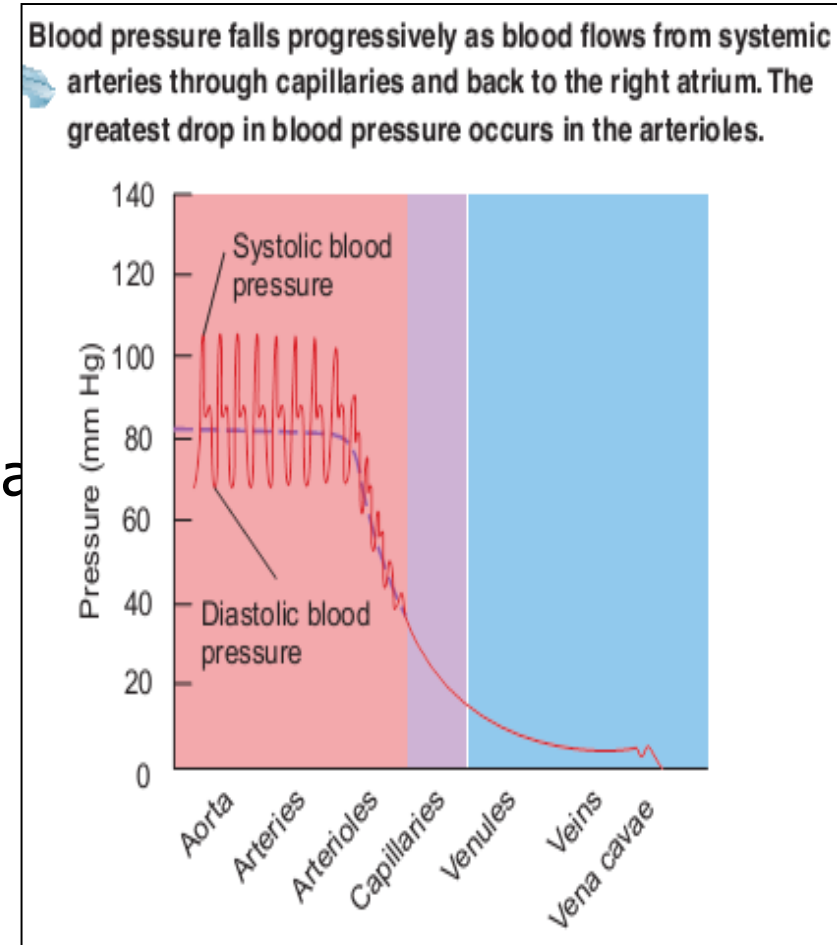
- the highest arterial pressure during a cardiac cycle (it is **120mmHg**)

❑ Diastolic pressure

- the lowest arterial pressure during a cardiac cycle (it is **80mmHg**)

❑ Pulse pressure

- the **difference** between the **systolic** and **diastolic** pressure



$$PP \text{ (mmHg)} = SBP \text{ (mmHg)} - DBP \text{ (mmHg)}$$

Arterial blood pressure...

□ Mean arterial pressure (MAP)

- takes account of pulsatile blood flow in the arteries, & is the **best measure of perfusion pressure** to an organ.

- $\text{MAP} = \text{Diastolic arterial pressure} + (\text{pulse pressure}/3)$ but in the **arterioles** it drops & is only about **37mmHg**.

Arterial blood pressure...

- The pressure drop by the time blood reaches the **capillaries** as there is a lot of surface area.
- **Low capillary pressure** with **high surface area** is a good thing because of 2 reasons:
 1. Capillaries are thin walled, so unable to withstand **high pressure**.
 2. Blood flow through capillaries will be slow giving more adequate time for **material exchange** to occur.

Hypertension

- Chronic resting **BP >140/90** is considered hypertension.
- **Baroreceptor reflex** **doesn't "fix"** the hypertension because, the baroreceptors **reset** to the chronically high values.
 - In other words, because those hypertensive pressures are somewhat constant, the baroreceptors **"think"** those pressures are the **correct normal values**.

Types of hypertension

- Hypertension may be **primary** (essential) or **secondary**.
- **90%** of hypertensive people have **essential hypertension**, in which **no underlying cause has been identified**.
- **10%** (2ndry hypertension) is due to **identifiable disorders** (known causes), such as:
 - excess rennin secretion by the kidneys
 - arteriosclerosis, or hyperthyroidism.

Types of hypertension...

1^{ry} hypertension

- Essential/idiopathic (90-95%).
- Cause is unknown, but there is a strong **genetic** tendency.
- Contributing factors include:
 - obesity, stress, smoking & excessive ingestion of salt.

2^{ry} hypertension

- 5-10%.
- Definite causes can be established in 4 categories:
 - Cardiovascular hypertension
 - Renal hypertension
 - Endocrine hypertension
 - Neurogenic hypertension

2^{ry} hypertension

Cardiovascular hypertension

→ due to chronically elevated **TPR** / total peripheral resistance (**atherosclerosis**).

Renal hypertension

→ either due to **occlusion of renal arteries** or unable to eliminate **salt load**
→ increase **blood volume** → hypertension.

2^{ry} hypertension...

Endocrine hypertension

- due to **endocrine disorder**
- **Pheochromocytoma** = adrenal tumor ↑ **epinephrine** & **norepinephrin**.
- **Conn's syndrome** =
↑ **aldosterone** due to hyper secreting cell.

Neurogenic hypertension

- occurs as a result of **neural lesions** due to
- defect in **cardiovascular control center** or **baroreceptors** or
- occurs as a compensatory response to **reduction in cerebral blood flow**.

Hypertension treatment

- Methods for **treating hypertension** include;
 - Weight loss.
 - Stress reduction.
 - Beta-blocking agents (reduce cardiac output).
 - Vasodilators:
(α -blocking agents, Ca^{++} channel blockers).
 - Diuretics (decrease blood volume)
 - Others.

Hypotension

- Blood pressure **<100/60 mmHg**.

low blood pressure



low blood flow to the tissues



damage of tissues



Circulatory shock

- ✓ Any condition in which blood vessels are **not filled adequately** & blood **cannot circulate normally**.

Circulatory Shock

- **Impairs tissue metabolism** because without proper circulation, **oxygen** & **nutrients** cannot get to the tissues & **wastes** cannot be taken away.
- There are five types of shock:
 - Hypovolemic shock
 - Cardiogenic shock
 - Neurogenic shock
 - Anaphylactic shock
 - Septic shock

Circulatory Shock...

1. Hypovolemic shock

- the most common type of shock
- results from a **large scale loss of blood** due to acute hemorrhage, vomiting, diarrhea, or extensive burns.
- **Signs** & events include
 - weak, rapid pulse
 - intense vasoconstriction
 - cold, clammy, cyanotic skin.

Circulatory Shock...

- The **pulse is rapid** because the heart tries to $\uparrow$ **CO** to offset the loss of blood volume.
- The **pulse is weak** because the $\downarrow$ in volume $\Rightarrow$ $\downarrow$ in venous return & thus a $\downarrow$ in stroke volume.
- A common **treatment** for shock is to **raise the legs** of the affected individual.
- Raising the legs will **increase venous return** and thus help move blood from the limbs to more integral areas like the **brain**.

Circulatory Shock...

2. Cardiogenic (vascular shock)

- where blood volume is normal but circulation is impaired due to **extensive vasodilation**.

3. Anaphylactic shock

- systemic allergic reaction with massive **histamine release**.

Circulatory Shock...

4. Neurogenic shock

- a **failure** to maintain **sympathetic vasomotor tone**.

.

5. Septic shock

- severe bacterial infection - **bacterial toxins** are often **potent vasodilators**.

Controlling Blood Pressure

- Blood pressure is subject to **local, neural** and **hormonal** controls.
- These BP controls are either for the short-term or for the long-term.
 - **Short-term control** is accomplished by affecting **cardiac output** (heart rate & stroke volume) and **peripheral resistance**.
 - **Long-term control** is accomplished via changes in **blood volume**.

Local Control of BP

- ❑ The ability of **tissues** to regulate their own blood supply is an example of **autoregulation**.
- ❑ If a tissue is inadequately perfused,
 - it can become **hypoxic** &
 - metabolites (**CO₂**, **K⁺**, **H⁺**, **lactate** & **adenosine**) can accumulate.
- ❑ The lack of oxygen and a high concentration of these metabolites stimulate **vasodilations**.

Local Control of BP...

- ❑ Increase in tissue metabolites & lack of oxygen indicates that blood flow to the tissue is **insufficient**.
- ❑ The metabolites **signal** the endothelium to release a **vasodilator** which will ↑ blood flow to the area.
- ❑ In response to a more chronic denial of oxygen, the deprived tissues can increase perfusion by stimulating angiogenesis - the growth of new blood vessels.

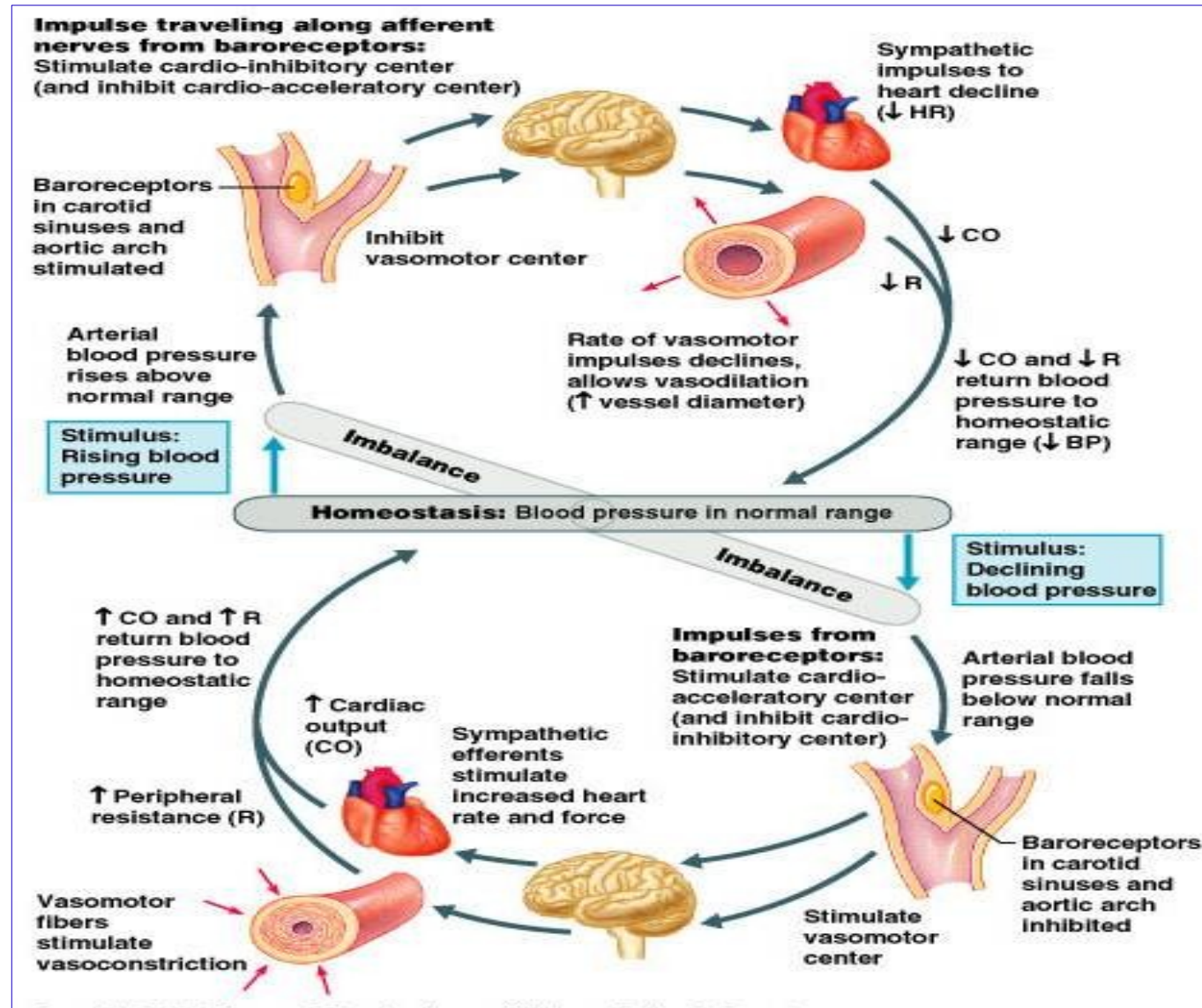
Neural Control of BP

- Located in the **medulla oblongata** is the **vasomotor center** which controls the degree of **sympathetic discharge** onto the arterioles & thus their vasomotor tone.
- In response to $\uparrow$ or $\downarrow$ in BP, **baroreceptors** will send information to the **vasomotor center**.
- Which will then $\uparrow$ or $\downarrow$ the degree of **arteriolar constriction** by adjusting the sympathetic discharge of **norepinephrine** onto the arterioles.

Neural Control of BP...

- The baroreceptors project to the **cardioinhibitory & cardioacceleratory centers** as well **adjust cardiac output** so as to achieve the desired change in BP.
- Signals from the baroreceptors to these medullary centers travel via the **glossopharyngeal** and **vagus nerves** (cranial nerves IX & X respectively).
- The **parasympathetic impulses** to the **heart** also travel via the **vagus nerve** while the **sympathetic impulses** to the heart travel via **sympathetic cardiac nerves**.

Neural Control of BP...



Neural Control of BP...

- **Chemoreceptors** (aortic bodies & carotid bodies) are neurons very **sensitive to** blood levels of **CO₂** & **H⁺**.
- In response to an **↑** in plasma **CO₂** or **H⁺ (↓pH)**, these neurons will signal the respiratory centers in medulla oblongata, which respond by **↑the rate & depth of respiration**.
- These neurons also project to the **cardiovascular centers**. Information from these chemoreceptors travels via the **glossopharyngeal & vagus nerves**.

Neural Control of BP...

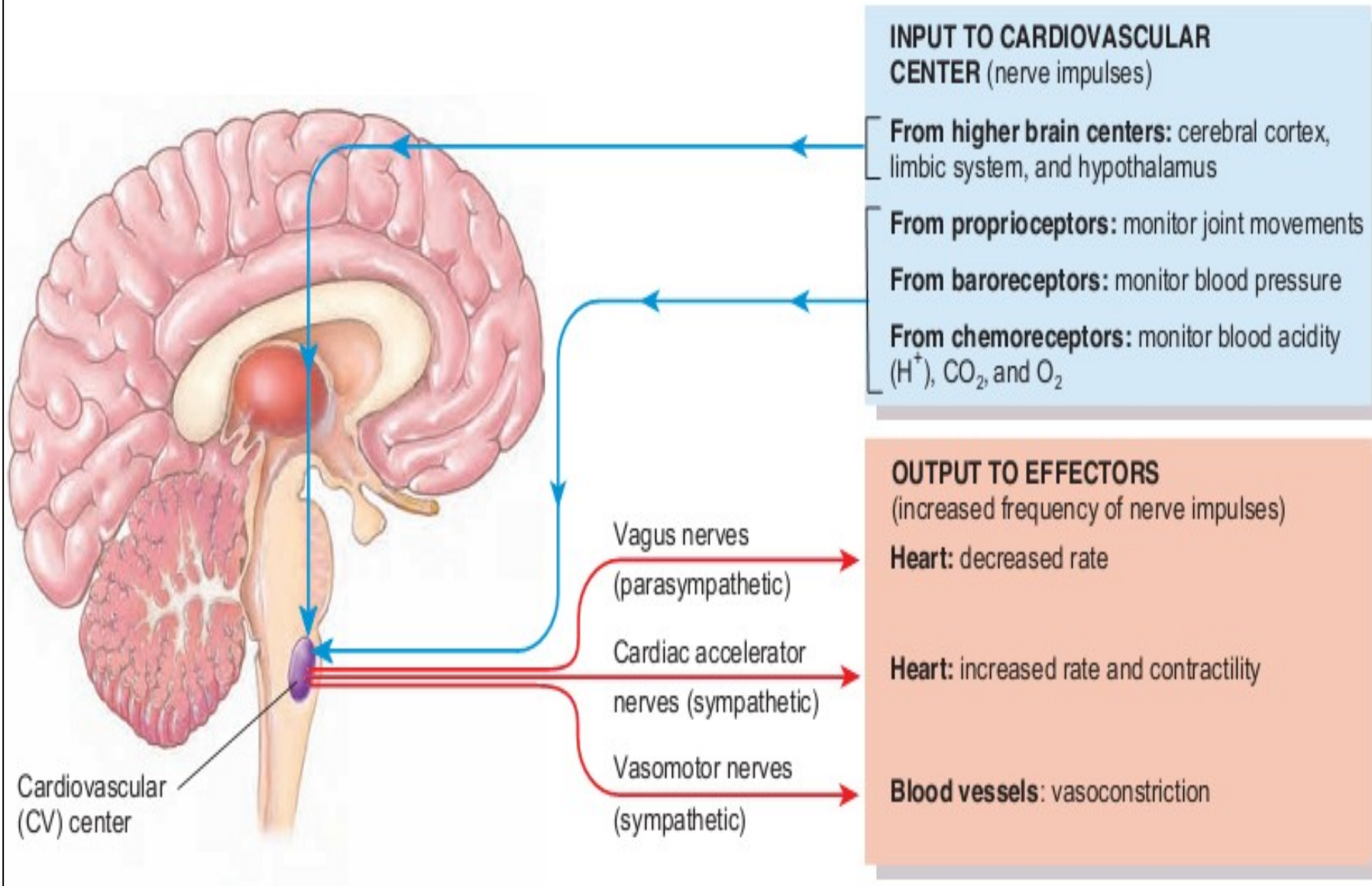
- They cause ↓cardioinhibitory activity,
↑**cardioacceleratory** & **vasomotor center activity**.
 - Thus the ↑CO₂ also stimulates ↑cardiac output.
- The idea is that not only do we want to ↑intake of O₂ & loss of CO₂ (via ↑respiratory rate & depth), we also want to speed up the transport of O₂ to the tissues & removal of CO₂ from the tissues.

Neural Control of BP...

- This is accomplished by **increasing cardiac output**.
- Higher brain centers such as the **hypothalamus** & **cerebral cortex** also influence BP by affecting the **cardiac** and **vasomotor centers**.
- **BP changes** are associated with **stressful, emotional situations** as well as during the **sexual response**.



The cardiovascular center is the main region for the nervous system regulation of heart rate, force of heart contractions, and vasodilation or vasoconstriction of blood vessels.



Hormonal Controls of BP

- There are various **hormones** that can affect BP.

Adrenal Medulla Hormones

- In response to **environmental stress**, the adrenal glands release **epinephrine** (85%) & **norepinephrine** (15%) into the bloodstream.
- Both bind to vascular smooth muscle & cause **vasoconstriction** this **↑peripheral resistance** & thus **blood pressure**.
- Both also bind to heart muscle & nodal tissue & **↑heart rate** & **stroke volume** = **↑cardiac output** & thus **blood pressure**.

Hormonal Controls of BP...

Antidiuretic Hormone (ADH)

- Produced by the hypothalamus & subsequently released via the posterior pituitary gland in response to;
 - **low BP** or **high plasma osmolarity**.
- As its name implies, it acts to **prevent diuresis** (urination) which helps **conserve blood volume** & **prevents further BP decline**.
- Also acts as a **potent vasoconstrictor**.

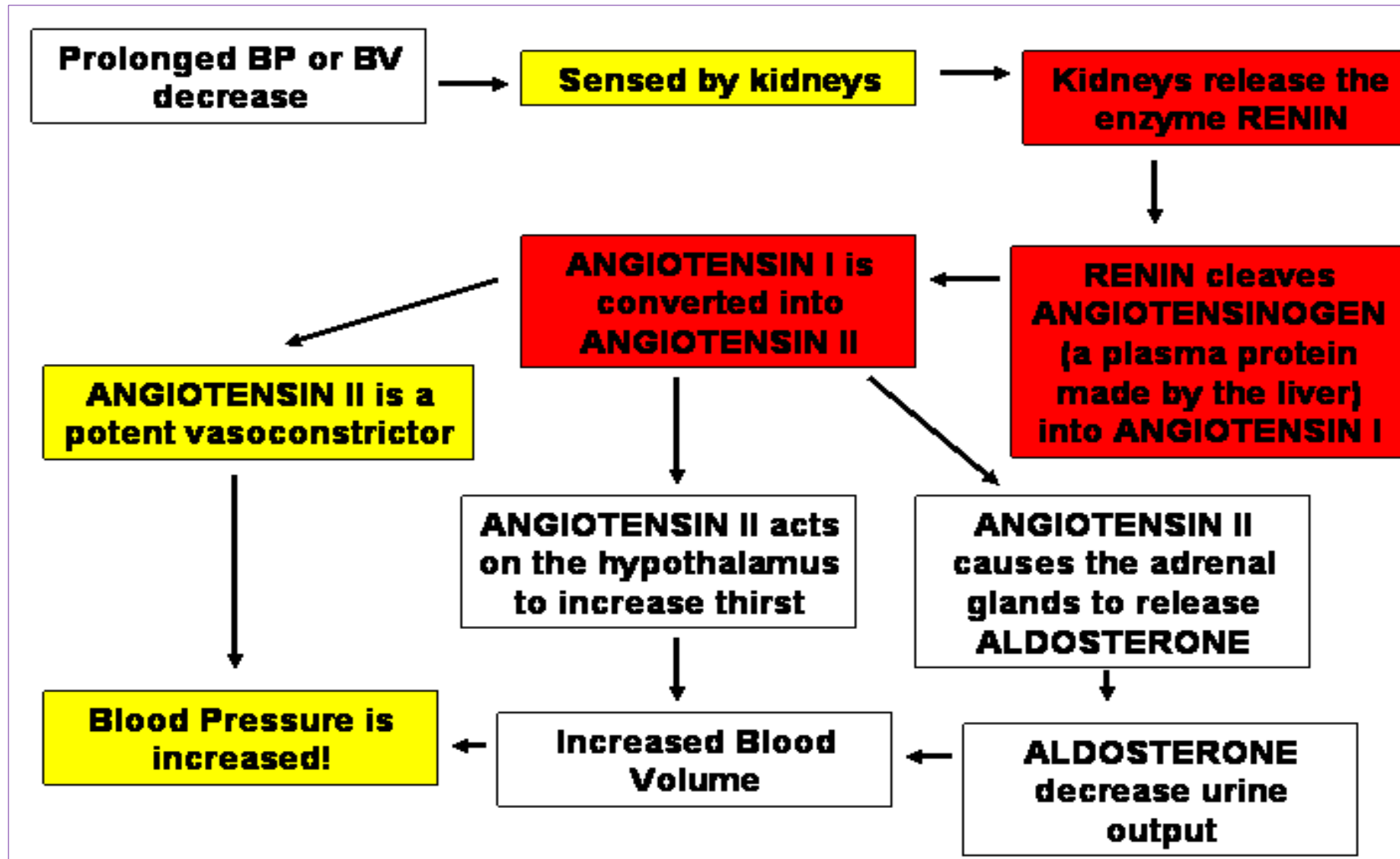
Long Term Blood Pressure Control

- Are mediated by the **kidneys** & alter **blood volume** rather than peripheral resistance.
- The kidneys regulate blood volume both directly & indirectly.
 1. **Direct mechanism**; as **blood volume** or **BP**↑, the rate at which fluid enters the kidney tubules ↑ & urine volume↑. The **excretion of more water in the urine** will ↓**blood volume** & thus ↓BP.
 2. **Indirect renal regulation** of blood pressure is slightly more complicated. It is also known as the **renin-angiotensin-aldosterone mechanism**. It works for a decrease in blood pressure.

Renin-Angiotensin-Aldosterone System

- ↓**BP** prompts the kidneys to release the **enzyme renin**.
- Renin begins the catalysis of a series of reactions that culminates with the production of a **protein** known as **angiotensin II**.
- Angiotensin II is a **potent vasoconstrictor**. Angiotensin II causes the secretion of the steroid **hormone aldosterone** from the adrenal cortex.
- Aldosterone causes a ↓**in urine output** (prevents any further ↓**in blood volume** by conserving water).
- Take a look at the entire mechanism.

RAAS...



Atherosclerosis

- Deposits of **fatty substances, cholesterol, wastes, calcium**... build up in the inner lining of an artery.
- The end-stage of this disease process is known as **arteriosclerosis**.
- The build-up is known as a **plaque**. The presence of a plaque will $\uparrow$ **resistance** & thus $\downarrow$ **blood flow**.
- The plaque will also **attract platelets** and thus increase the likelihood of **coagulation**.

Capillary Exchange

- Capillaries are designed for exchange of **nutrients, gases, wastes, & signaling molecules**.
- All diffuse between capillary plasma and interstitial fluid depending on their **concentration gradient**.
- The **nature of the substance** being exchanged is also important.
 - **Hydrophobic molecules** can diffuse directly through the capillary endothelial cells, whereas
 - **Hydrophilic molecules** must diffuse through either the intercellular clefts or capillary fenestrations.

Capillary Exchange...

- As nutrients, wastes, & gases are being exchanged, **bulk fluid flow** is also occurring.
- Fluid is forced out of the **capillaries** at the **arterial end**, but most of it returns at the **venous end**.
- This bulk flow helps determine the relative distribution of fluid in the extra cellular space, i.e.,
 - what % of ECF is found within plasma and what % is within the tissues (interstitial space).

Capillary Exchange...

- The direction and volume of fluid that flows across the capillary walls reflects the balance between 4 forces:
 1. Hydrostatic Pressure (HP) within the Capillary
 2. Hydrostatic Pressure of the Interstitial Fluid
 3. Plasma Colloid Osmotic Pressure
 4. Interstitial Space Fluid (ISF) Colloid Osmotic Pressure

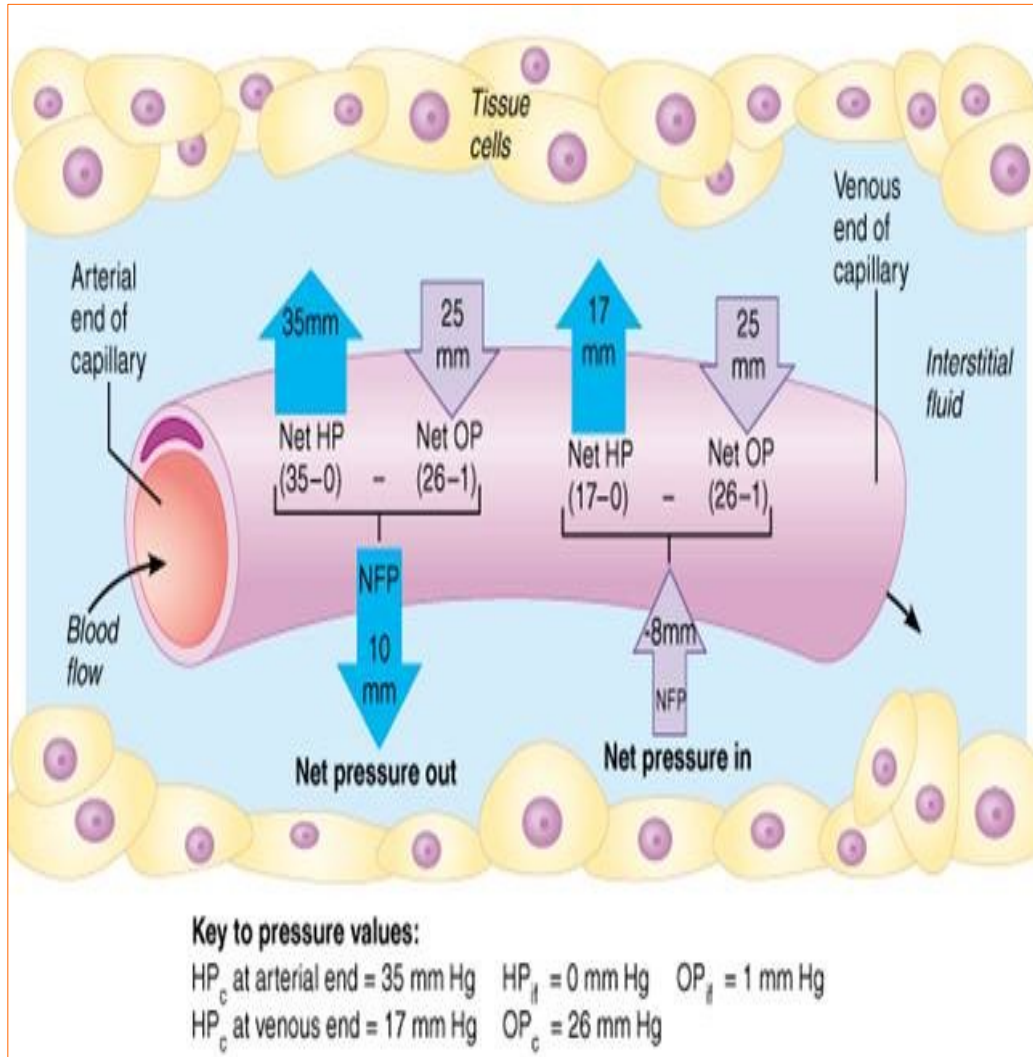
Capillary Exchange...

- Capillary hydrostatic pressure & ISF osmotic pressure
 - 2 forces that move **fluid out of the capillaries**
- Capillary osmotic pressure & ISF hydrostatic pressure
 - 2 forces that promote **fluid** movement **into capillaries**
- At the arterial end of the capillary, we have 10mmHg of pressure pushing **fluid out** of the capillary into the interstitial space.
- At the venous end of the capillary, we have 8mmHg of pressure **pulling fluid** back into the capillary from the interstitial space.

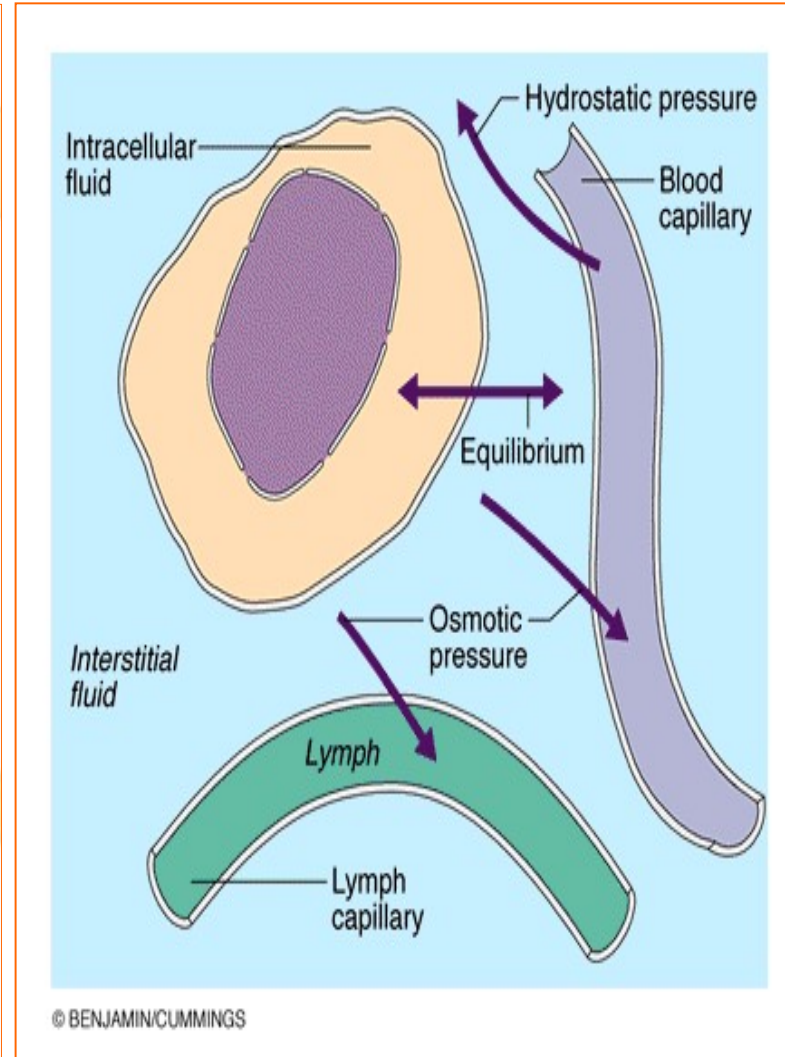
Capillary Exchange...

- Notice that the filtration (**fluid loss**) at the arterial end of the capillary is **NOT balanced** by the reabsorption (**fluid gained**) at the venous end of the capillary.
- We have slightly **more fluid being forced out** at the arterial end of the capillary **than** **being forced in** at the venous end of the capillary.
- We actually lose about **3L/day** of fluid from the bloodstream into the interstitial space because of capillary filtration.
- Luckily our **lymphatic system** returns that fluid to the bloodstream.
- Look at this figure depicting the whole process.

Capillary Exchange



Fluid Movement



Factors affecting the formation of interstitial fluid

1. High Arterial Pressure = $\uparrow$ capillary HP $\uparrow$ ISF
2. Allergic Reaction = Histamine $\uparrow$ blood flow
 $\uparrow$ capillary HP $\uparrow$ ISF
3. Starvation = $\downarrow$ protein intake $\downarrow$ capillary OP $\uparrow$ ISF
4. Capillary injury (toxins, severe burns, crush injuries) = $\uparrow$ capillary permeability $\uparrow$ ISF

Lymphatic System

- The body system responsible for adaptive **immunity** and some aspects of innate immunity.
- Consists of :
 - **lymph, lymphatic vessels**, a number of structures & organs containing **lymphatic tissue** & **red bone marrow**.
- Lymphatic tissue is a specialized form of reticular connective tissue that contains large numbers of **lymphocytes**.

Lymphatic System...

- Most components of blood plasma filter out of blood capillary walls to form interstitial fluid , the fluid that surrounds the cells of body tissues.
- After interstitial fluid passes into lymphatic vessels, it is called **lymph (clear fluid)**.
- Both fluids are chemically similar to blood plasma.
 - The main difference is that interstitial fluid and lymph contain less protein than blood plasma because most plasma protein molecules are too large to filter through the capillary wall.

Lymphatic System...

- Each day, about **20 liters** of fluid **filter** from blood into tissue spaces.
- This fluid must be returned to the cardiovascular system to maintain normal blood volume.
- About **17 liters** of the fluid filtered daily from the arterial end of blood capillaries return to the blood directly by **reabsorption** at the venous end of the capillaries.
- The remaining **3 liters** per day pass first into lymphatic vessels & are then returned to the blood.

Functions of the lymphatic system

1. Drains excess interstitial fluid.

- Lymphatic vessels drain excess interstitial fluid & leaked proteins from tissue spaces & return them to the blood.
- This activity helps maintain fluid balance in the body & prevents depletion of vital plasma proteins.

2. Transports dietary lipids.

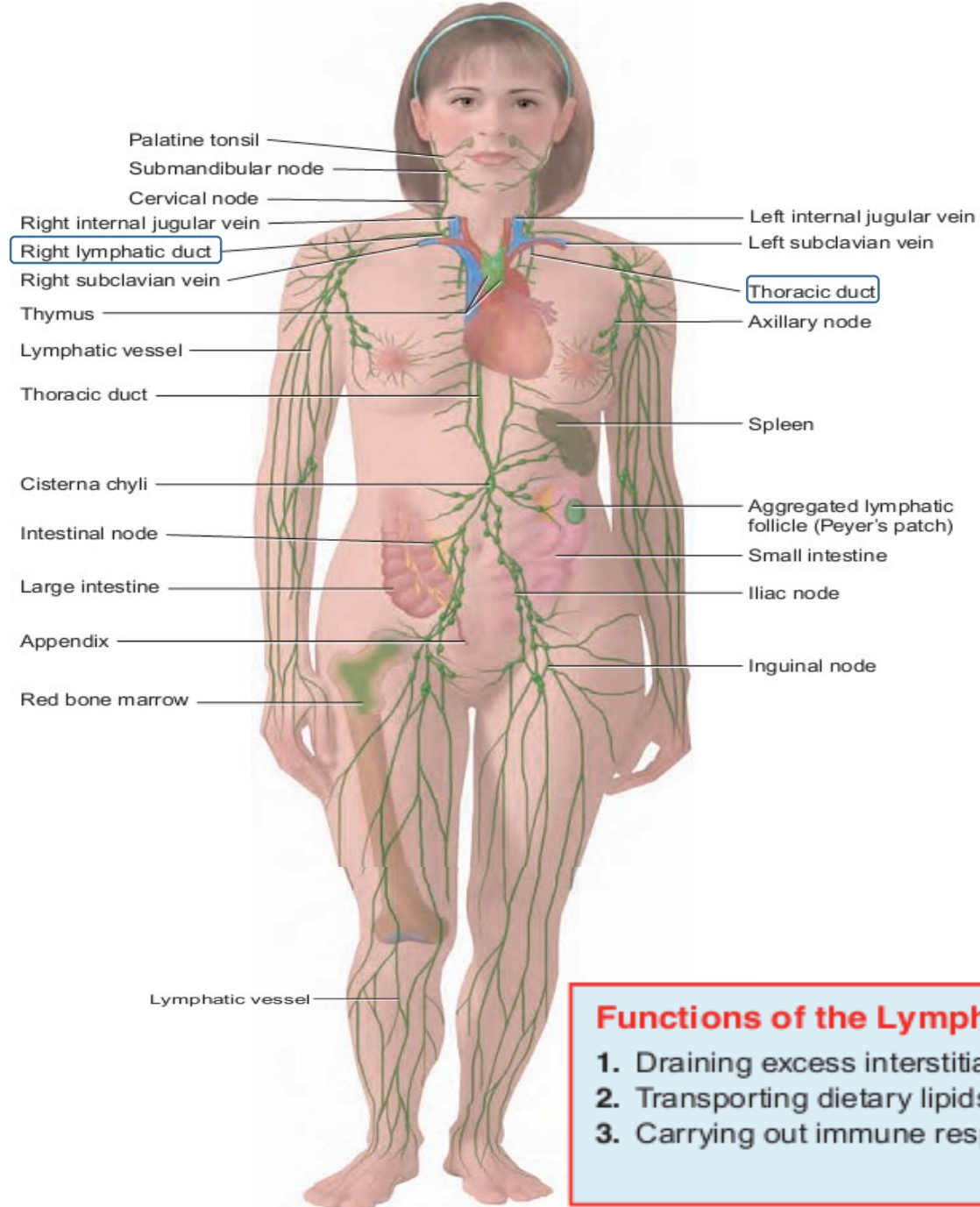
- Lymphatic vessels transport the lipids & lipid-soluble vitamins (A, D, E, K) absorbed by the gastrointestinal tract into the blood.

3. Carries out immune responses.

- Lymphatic tissue initiates highly specific responses directed against particular microbes or abnormal cells.

Lymphatic Organs and Tissues

- Lymphatic organs & tissues, which are widely distributed throughout the body, are classified into two groups based on their functions.
- **Primary lymphatic organs**
 - sites where stem cells divide & develop into *mature B cells & T cells*, include the **red bone marrow** (in flat bones & the ends of the long bones of adults) & **thymus**
- **Secondary lymphatic organs and tissues**
 - the sites where most immune responses occur, include **lymph nodes** , **spleen** and **lymphatic nodules**.



Anterior view of principal components of lymphatic system



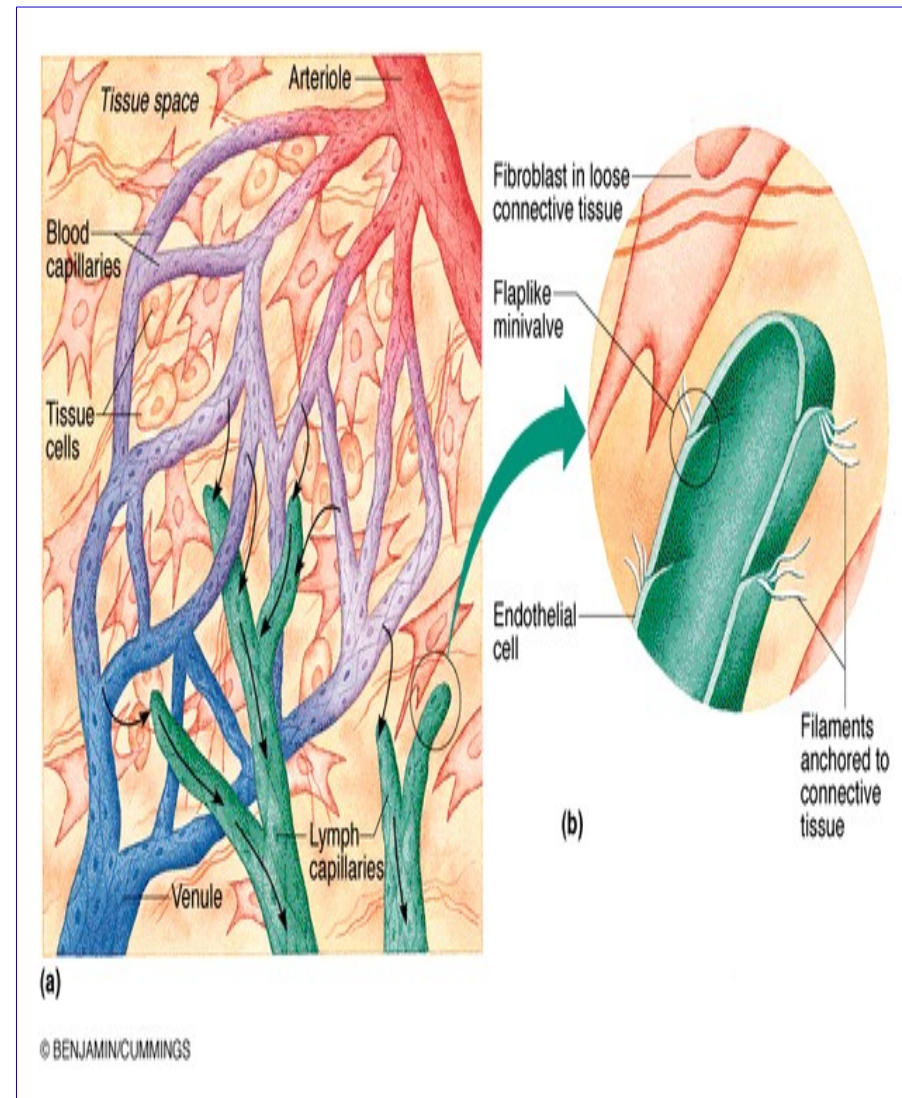
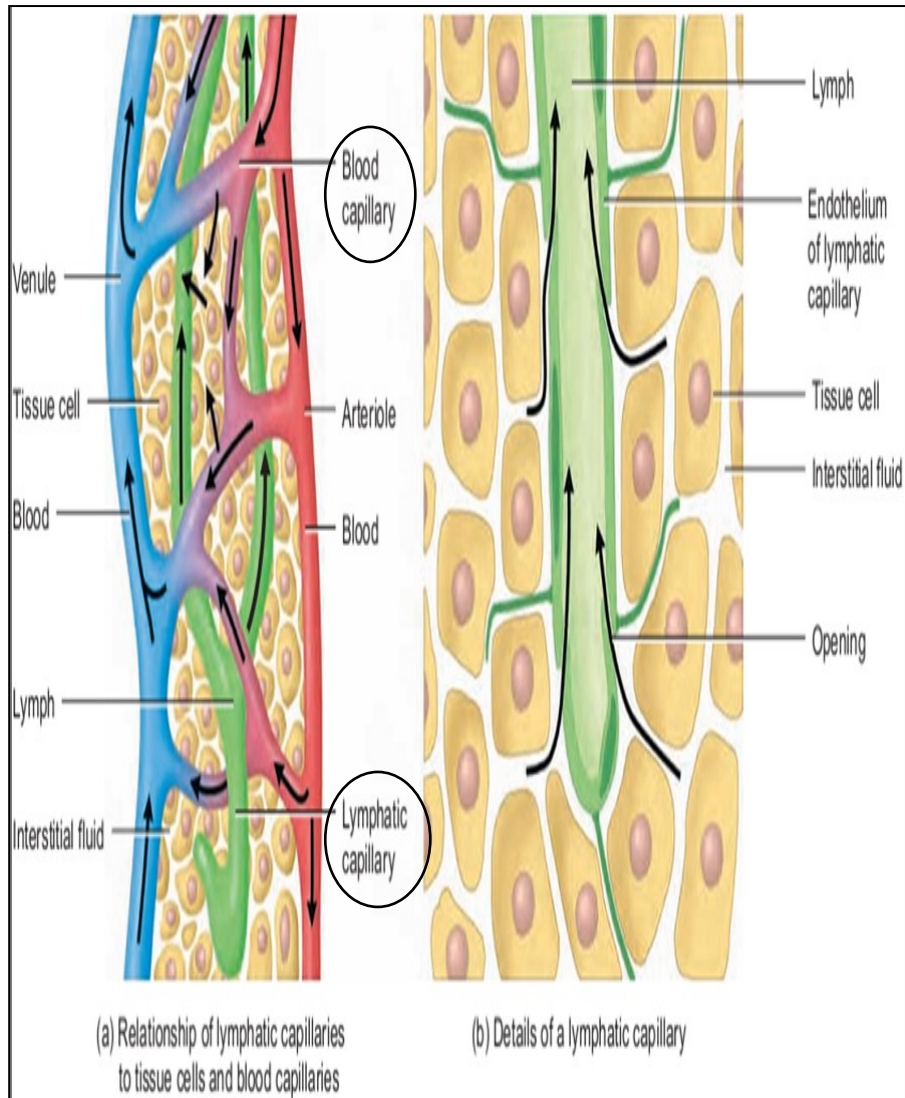
(b) Areas drained by right lymphatic and thoracic ducts

- Area drained by right lymphatic duct
- Area drained by thoracic duct

Functions of the Lymphatic System

1. Draining excess interstitial fluid.
2. Transporting dietary lipids and lipid-soluble vitamins to the blood.
3. Carrying out immune responses.

Lymphatic capillaries



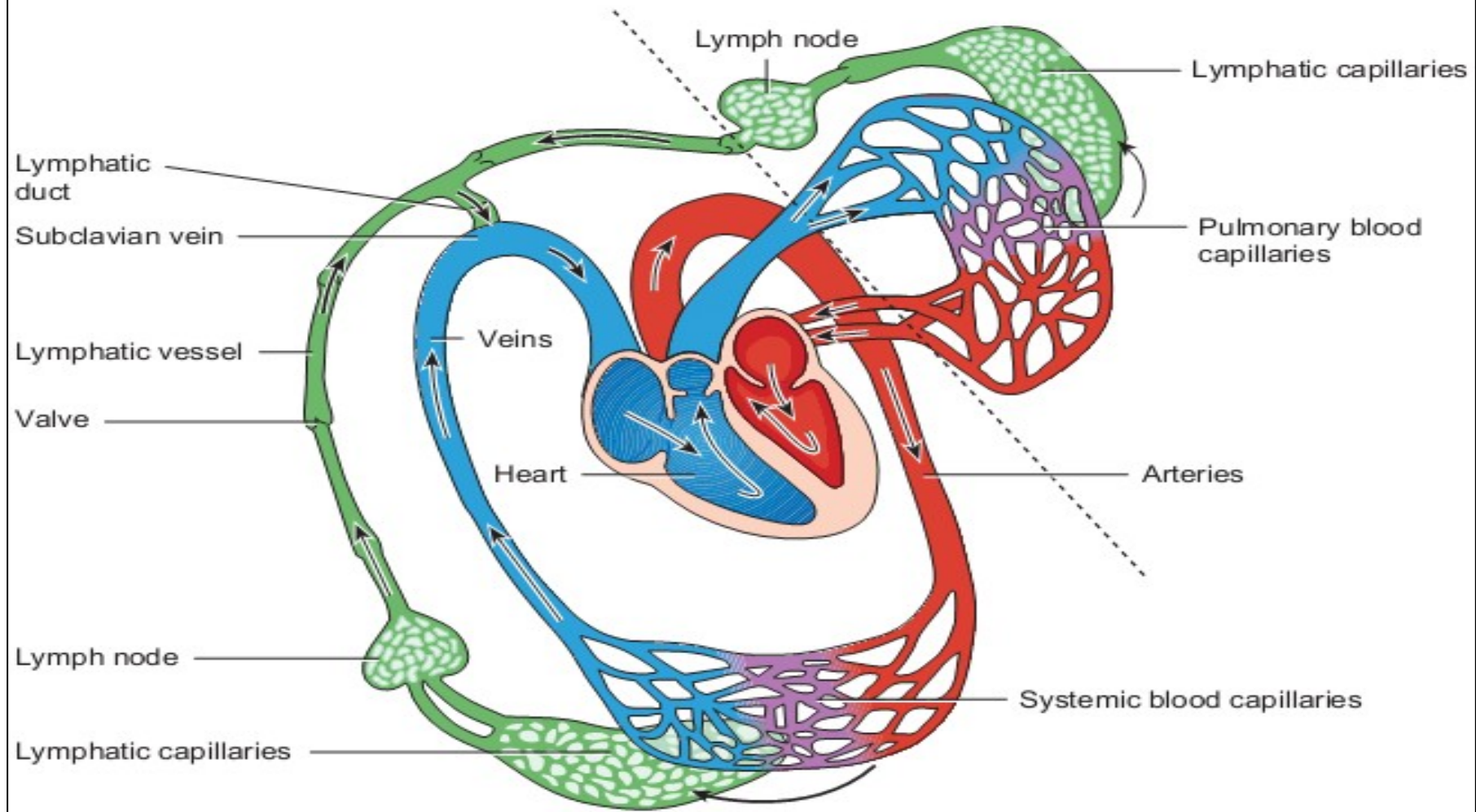
Relationship of lymphatic vessels and lymph nodes to the cardiovascular system. Arrows show the direction of flow of lymph and blood.



The sequence of fluid flow is: blood capillaries (blood plasma) → interstitial spaces (interstitial fluid) → lymphatic capillaries (lymph) → lymphatic vessels and lymph nodes (lymph) → lymphatic ducts (lymph) → junction of jugular and subclavian veins (blood plasma).

SYSTEMIC CIRCULATION

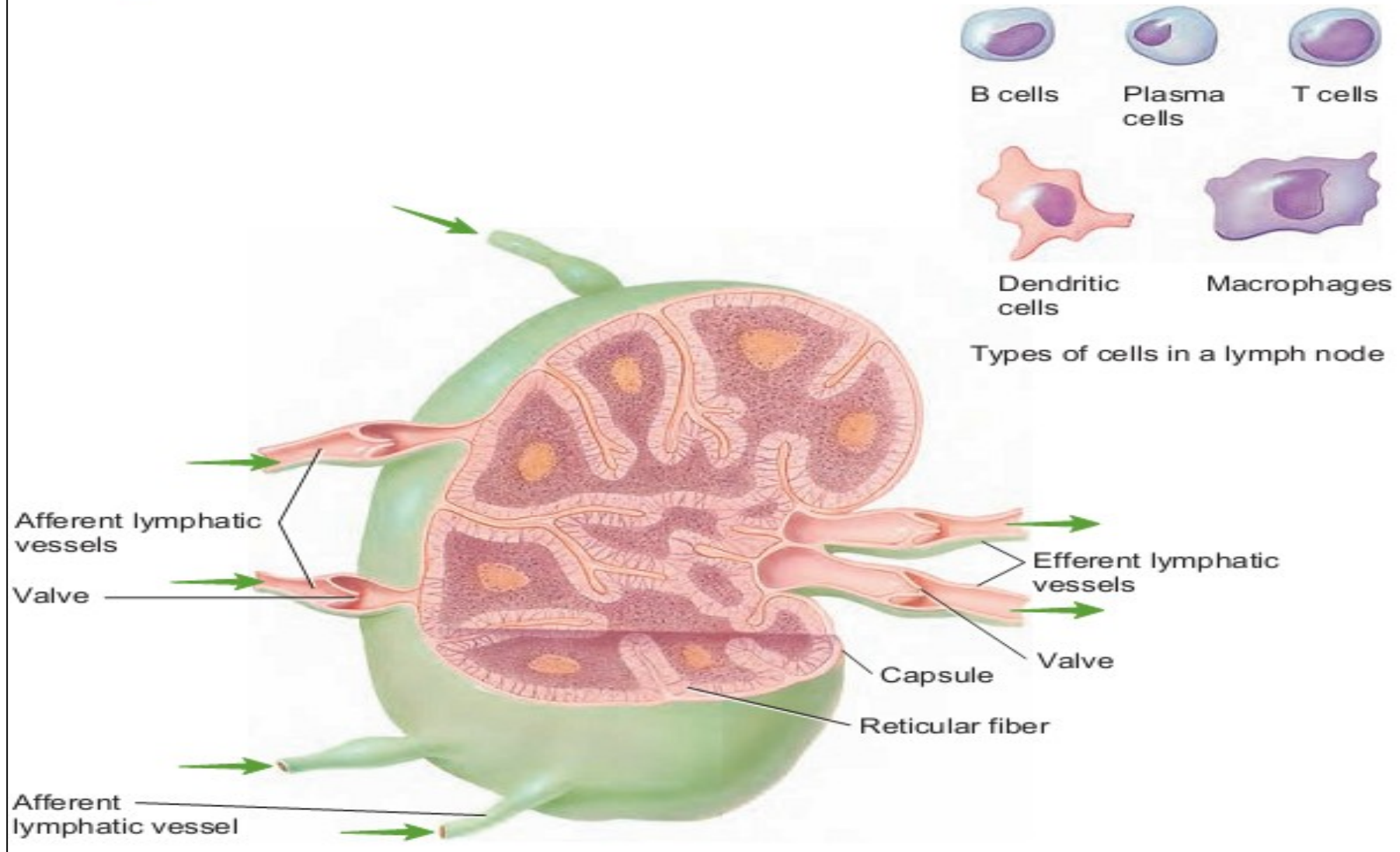
PULMONARY CIRCULATION



Structure of a lymph node (partially sectioned). Green arrows indicate direction of lymph flow into and out of the lymph node.



Lymph nodes are present throughout the body, usually clustered in groups.



Lymphatic System...

Lymph

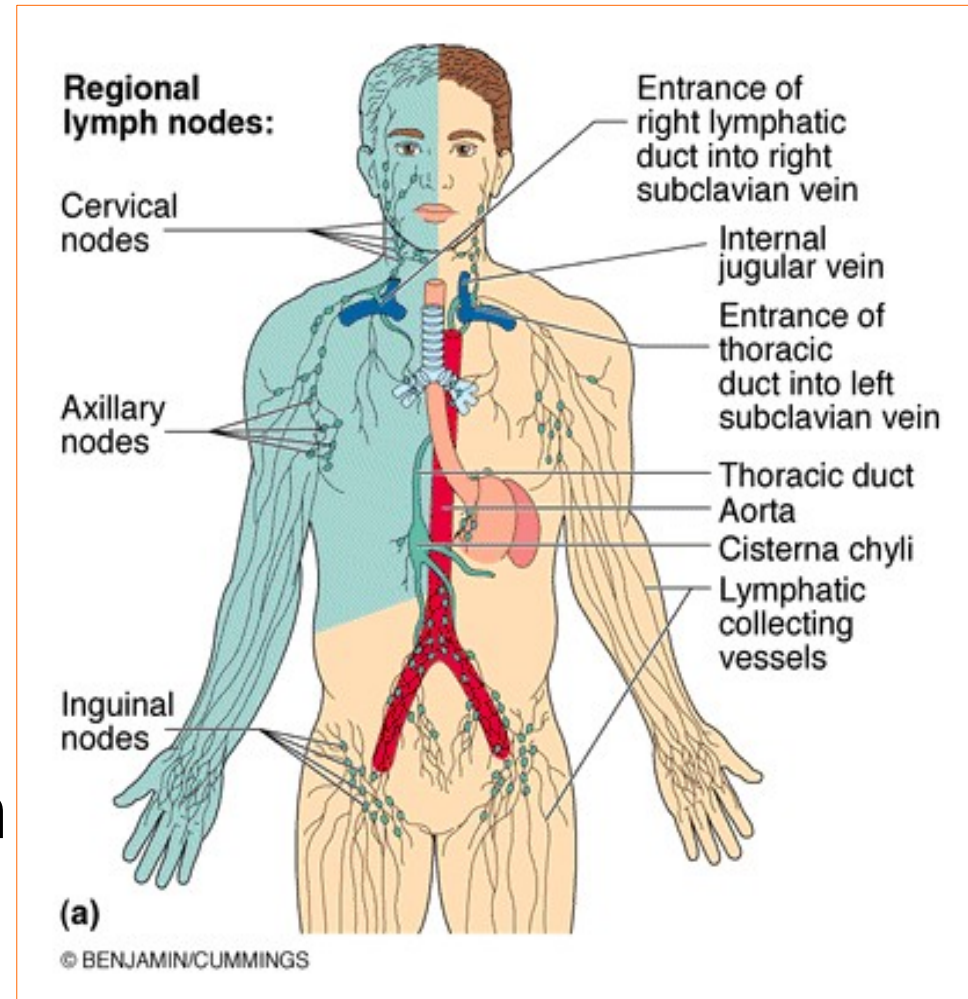
- Fluid of the lymphatic system
- Similar to blood plasma and interstitial fluid

Lymphatic Vessels

- Transport lymph
- Lymph is returned to the circulatory system at either;
 - the right or left **subclavian veins**

Lymph Nodes

- Numerous in the neck, axillary and inguinal regions of the body
- Filter lymph
 - Microorganisms
 - Cancer cells
- Lymphocytes and Monocytes are found in the lymph nodes.



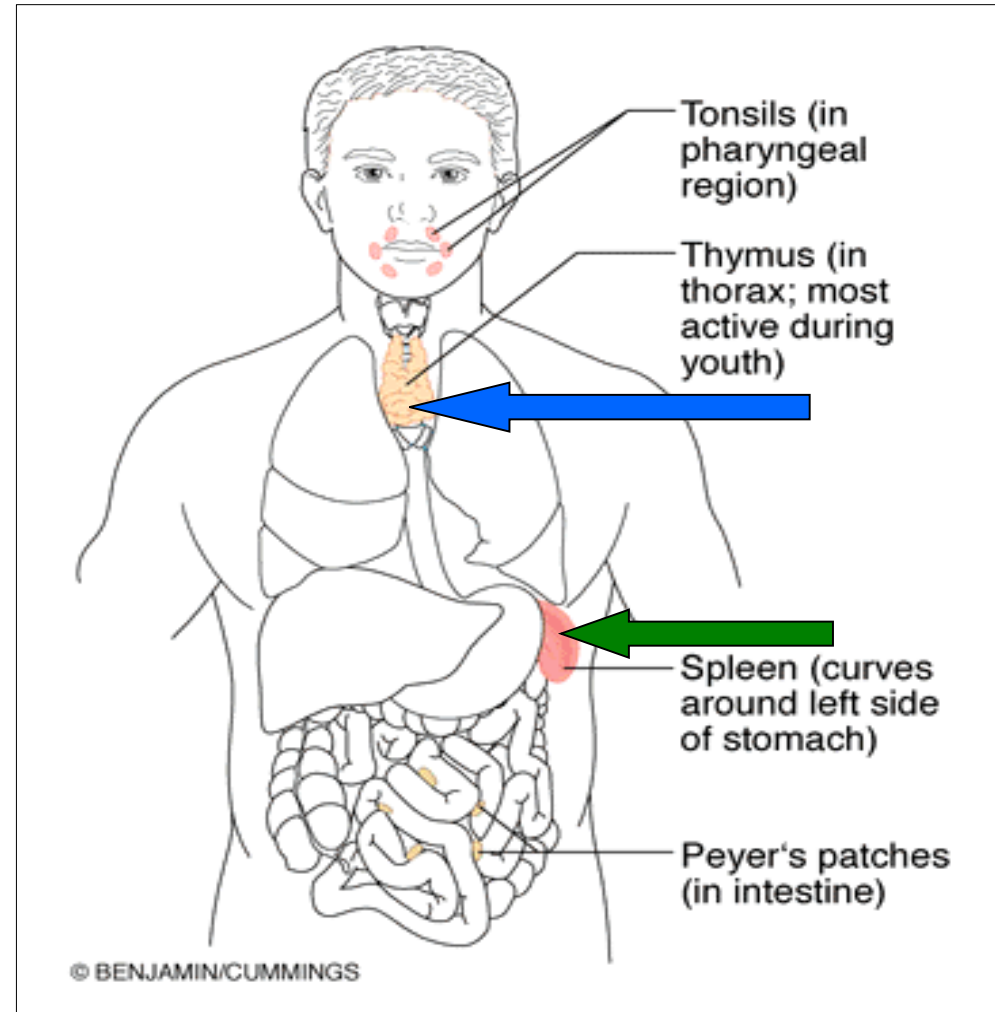
Lymphatic organs

Thymus

- Programs some lymphocytes to develop into T-cells

Spleen

- Filters blood
 - Worn out RBC
 - Bacteria
- Contain Lymphocytes and Monocytes



Edema

- An abnormal accumulation of interstitial fluid.

Causes

1. Blockage of lymphatic system
2. Increased pressure in veins
3. Lack of albumin
 - Decreases fluid returning to blood capillaries by osmosis
4. Inflammation

Interesting Facts

- Heart Disease is the number one killer in American women.
- 16.7 million deaths are result forms of cardiovascular disease, heart disease and stroke.
- Risk factors of developing heart disease :
 - Stress
 - Eating high fat foods
 - Obesity
 - Tobacco and
 - Alcohol use

Interesting Facts...

- Recent research suggests that taking a small dose of aspirin daily may help prevent a heart attack (because aspirin inhibits platelet clumping).
- The length of all your blood vessels lined up is about **60,000 miles long!**
 - To put this in perspective, the **Earth's circumference** is **40,075.02 km** & 60,000 miles is around **96,000 km** - so your blood vessels would go **twice around the world** and still have some to spare!

Ways to a Healthy Heart

- Eating healthy, good nutrition.
- Fitness and Exercise.
- Having a healthy lifestyle; don't drink, smoke, or do drugs.
- Lowering LDL cholesterol and high blood pressure.
- Reduce the fat, sodium, and calories in your diet.